

THE CONSTANTS OF THE FUNCTIONAL EQUATIONS OF *L*-FUNCTIONS

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Introduction

The essential results of this article are the following.

(A) We give a simple proof of Langlands' theorem [9], already proved up to sign by Dwork [6], which makes it possible to write the constant in the functional equations of Artin L -functions, or of Weil L -functions [19], as a product of local constants.

The theorem to be proved is local, and Langlands' proof had the elegance of being purely local. Essentially, the problem is to prove a multiplicative identity between somewhat generalized Gauss sums whenever one has a $\mathbb{Z}$ -linear relation between characters of local Galois groups induced by characters of one-dimensional representations of subgroups. Langlands constructed a generating set of such relations, and reduced the identities to be proved to identities between Gauss sums, which he proved using Stickelberger's theorem.

The proof presented here is very close to the proof [5] of the Hasse–Davenport identity (cf. also Weil [18]). It uses a global argument (4.12), and the fact that the local constants to be defined are essentially trivial at infinity, in the unramified case, and in certain very ramified cases (4.14).

(B) Let K be a function field and

$$\rho : \text{Gal}(\overline{K}/K) \longrightarrow \text{GL}(n, \mathbb{Q}_\ell)$$

an ℓ -adic representation of the Galois group of K , unramified almost everywhere. Grothendieck showed that the Euler product over the places of K , with p^{-s} replaced by an indeterminate t ,

$$Z(\rho, t) = \prod_v \det(1 - F_v t^{\deg(v)}, (\mathbb{Q}_\ell^n)^{\rho(I_v)})^{-1} \in \mathbb{Q}_\ell[[t]],$$

is a rational function, and that a functional equation relates $Z(\rho, t)$ to

$$Z^\vee(\rho, t) : \quad Z^\vee(\rho, p^{-1}t^{-1}) = \varepsilon Z(\rho, t),$$

where the constant ε is a monomial at^b . This theory is summarized in §10. Result (A) suggests a formula expressing ε as a product of local constants. We prove that this formula is true when ρ belongs to an infinite compatible system of ℓ -adic representations (9.3 and 9.9). According to Weil [16] and Jacquet–Langlands [8], this formula, applied to the compatible system of ℓ -adic representations defined by an elliptic curve over K , makes it possible to attach to every elliptic curve over K a modular form for $\text{GL}(2, K)$.

It would be very interesting to know whether the formula obtained remains valid for an arbitrary ℓ -adic representation.

Here is the content of the various sections, and their interdependence.

The beginning of §1 gathers a few results on induced representations of finite groups which will be useful to us.

The end of §1, from 1.11 to the end, is not used in the rest of the article. For a solvable group G , it describes a set R of $\mathbb{Z}$ -linear relations between the characters of G induced by one-dimensional characters of subgroups, and proves that every such relation is a linear combination of elements of R . The result and its proof are implicitly contained in [9].

Sections 2 and 3 serve mainly to fix notation. In §2 we recall the structure of the Galois group of a local field [10], define the Weil groups, a variant of Galois groups, and give a few statements from class field theory. The reader should note that throughout this article the isomorphism of local class field theory sends a uniformizer to the inverse of the Frobenius substitution; the reason is given in 3.6. Section 3 recalls Tate's thesis and Artin [1].

Section 4 contains the proof of result (A) above. The formula sheet in §5 gives a list of fundamental properties of the local constants constructed in §4, and a few identities between Gauss sums that follow from them.

Sections 6, 7, and 8, the latter depending only on §2, are preliminaries to the proof of (B), given in §9. In the statements of §§3 and 4, we were very careful to use only rational formulas, involving neither passage to a limit nor square roots, even of a positive integer. We harvest the benefit of this in §§6 and 7. In §6, for K a non-archimedean local field and V a modular representation of $W(\overline{K}/K)$ over a field A of characteristic $\ell \neq p$, we define, by transport of structure for $\ell = 0$ and by reduction modulo ℓ for $\ell \neq 0$, a local constant

$$\varepsilon(V, \psi, dx) \in A^*.$$

In §7, for K global of characteristic p and V a modular representation of $W(\overline{K}/K)$, we prove, by reduction modulo ℓ , a functional equation for the corresponding L -function $L \in \Lambda(t)$.

In §8 we give the isomorphism classes of ℓ -adic representations of the Weil group of a local field a purely algebraic description, independent of the topology of $\mathbb{Q}_\ell$. This is used to define compatibility of ℓ -adic and ℓ' -adic representations. The notion introduced is finer than Serre's [12].

Section 9 gives the proof of (B) (9.3 and 9.9). The idea is that, to prove an identity, it is enough to prove it modulo ℓ for infinitely many ℓ . We also give, for function fields, a partial answer to a question of Serre by showing (9.8) that two ℓ - and ℓ' -adic representations giving the same characteristic polynomial of Frobenius, assumed to lie in $\mathbb{Q}[t]$, at almost all places satisfy compatibility at the residual places.

Section 10 summarizes Grothendieck's theory of L -functions (cf. [7] and SGA 5), which is used crucially in §9. In 10.11 I try to explain the point of his method. The odd results 10.13 follow immediately from what precedes. I do not know their meaning.

§1. Virtual representations of a finite group

1.1. Let K be a commutative field. The most important case for us will be that in which K is algebraically closed of characteristic 0, for example $K = \mathbb{C}$. Let G be a group. We shall write $R_K(G)$ for the Grothendieck group of the category of finite-dimensional $K[G]$ -modules over K . It is identified with the free $\mathbb{Z}$ -module with basis the set of isomorphism classes of irreducible K -linear representations of G . It is a commutative ring with unit, for tensor product over K , and even a λ -ring in the sense of Grothendieck. Its elements are the virtual representations of G over K .

To define a homomorphism f from $R_K(G)$ to a commutative group X , it is enough:

- a) to define $f(V)$ for V a representation of G over K ;
- b) to check that for every exact sequence of representations

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0,$$

$$\text{one has } f(V) = f(V') + f(V'').$$

Applying this rule, one defines the dimension of a virtual representation

$$\dim : R_K(G) \longrightarrow \mathbb{Z},$$

and its determinant

$$\det : R_K(G) \longrightarrow \text{Hom}(G, K^*).$$

If H is a subgroup of finite index in G , define induction

$$\text{Ind}_H^G : R_K(H) \longrightarrow R_K(G)$$

as corresponding to extension of scalars

$$V \longmapsto K[G] \otimes_{K[H]} V.$$

This is an exact functor, and therefore passes to virtual representations. For $f : G \rightarrow H$, define restriction

$$\text{Res}_G^H : R_K(H) \longrightarrow R_K(G)$$

as restriction of scalars from $K[H]$ to $K[G]$. For f surjective one sometimes speaks rather of inflation.

For every group G , write G^{ab} for the largest abelian quotient of G . A K -quasi-character χ of G is a homomorphism

$$\chi : G \longrightarrow K^*;$$

we identify it with the K -quasi-character of G^{ab} through which it factors, and denote by $[\chi]$ the corresponding one-dimensional representation. For G finite, we shall use the words character and quasi-character interchangeably. The word character will never mean the character of a representation of dimension > 1 .

The following proposition is due to P. Gallagher [20].

Proposition 1.2. Let G be a group, H a subgroup of finite index, and

$$t : G^{\text{ab}} \longrightarrow H^{\text{ab}}$$

the transfer. For ρ a virtual representation of dimension 0 of H , and $x \in G^{\text{ab}}$, one has

$$\det(\text{Ind}_H^G(\rho))(x) = \det(\rho)(t(x)).$$

We shall prove, in only slightly greater generality, that if ε is the determinant of the permutation representation of G on G/H ,

$$\varepsilon : G \longrightarrow \mathfrak{S}_{G/H} \longrightarrow \{\pm 1\},$$

then, for ρ of arbitrary dimension,

$$\det(\text{Ind}_H^G(\rho))(x) = \varepsilon^{\dim \rho} \det(\rho)(t(x)).$$

This is fairly evident in topological terms. Let BG be the classifying space of G , and let B_H be that of H , regarded as a covering of BG . Representations of G are identified with local systems of finite-dimensional K -vector spaces on B_G , and Ind_H^G with the functor f_* . The transfer $H^1(B_G, X) \rightarrow H^1(B_H, X)$, for $X = K^*$, corresponds to the norm $f_!$. If $\rho \in H^1(B_G, K^*)$ is the class of a rank-one local system of K -vector spaces, then $f_!(\rho)$ is the class of $N(L)$, the local system on B_G which, locally on B_G , is the tensor product of the inverse images of L by the $[G : H]$ disjoint sections.

Put

$$\det V = \bigwedge^{\dim V} V$$

for a local system V , and let ε be the local system $\det f_* K$. Proposition 1.2 follows from the existence of a canonical isomorphism, for V a local system on B_H ,

$$\det f_* V \simeq \varepsilon^{\otimes \dim V} \otimes N(\det V).$$

The construction of this isomorphism is local on B_G , and reduces to the following construction.

Construction 1.3. For $(V_i)_{i \in I}$ a finite family of K -vector spaces of dimension d , and $\varepsilon = \det(K^I)$, one has canonically

$$\det\left(\bigoplus_{i \in I} V_i\right) \simeq \varepsilon^{\otimes d} \otimes \bigotimes_{i \in I} \det(V_i).$$

For $i_1, \dots, i_N$ the sequence of elements of I , the map is

$$\begin{aligned} & (e_{i_1,1} \wedge \dots \wedge e_{i_1,d}) \wedge \dots \wedge (e_{i_N,1} \wedge \dots \wedge e_{i_N,d}) \\ & \longmapsto (i_1 \wedge \dots \wedge i_N)^{\otimes d} \otimes (e_{i_1,1} \wedge \dots \wedge e_{i_1,d}) \otimes \dots \otimes (e_{i_N,1} \wedge \dots \wedge e_{i_N,d}). \end{aligned}$$

Parenthesis 1.4. Instead of a classifying space BG , one could more intrinsically have used, in the preceding argument, Grothendieck's classifying topos B_G (SGA 4, IV, 2.4, p. 315, for $E = (\text{Ens})$).

Assume G finite. We shall say that K is large enough if it contains all roots of unity whose orders divide the l.c.m. of the orders of elements of G . A group H will be called p -elementary if it is the product of a p -group and a cyclic group; elementary if it is p -elementary for some prime p . We shall need the following variant of Brauer's theorem on induced characters, obtained by omitting everywhere "of dimension zero."

Proposition 1.5. If K is large enough, every virtual representation of dimension 0 of G is a sum of virtual representations

$$\text{Ind}_H^G(\chi),$$

where χ is virtual of dimension 0, H is elementary, and χ is a linear combination of one-dimensional representations of H .

By Brauer's theorem applied to the trivial representation 1_G of G , there is a decomposition

$$1_G = \sum_H \text{Ind}_H^G(z_H), \quad H \text{ elementary.}$$

For y a virtual representation of dimension 0 of G , one then has

$$y = \sum_H y \cdot \text{Ind}_H^G(z_H) = \sum_H \text{Ind}_H^G(\text{Res}_H^G(y) \cdot z_H),$$

with $\dim(\text{Res}_H^G(y) \cdot z_H) = 0$. This reduces us to the case where G is elementary.

By linearity, one may suppose that y is of the form

$$\rho - \dim(\rho) 1_G,$$

with ρ irreducible, hence, since G is nilpotent, induced by a character χ of a subgroup H of G containing the center Z . Then

$$\rho - \dim(\rho) 1_G = \text{Ind}_H^G(\chi - 1_H) + (\text{Ind}_H^G(1_H) - [G : H] 1_G).$$

The first term is of the desired type, and the second is the inflation of a virtual representation of dimension 0 of G/Z . One concludes by induction on the order of G , noting that $|G/Z| < |G|$ if $G \neq \{e\}$.

Scholium 1.6. A homomorphism

$$f : R_K(G) \longrightarrow X$$

is known once its value on 1_G and on the

$$\text{Ind}_H^G(\chi - 1_H),$$

for H elementary and χ a character of H , are known.

1.7. Let A be a discrete valuation ring of unequal characteristic p , with fraction field K and residue field $k = A/\mathfrak{m}$. Recall the definition of the decomposition homomorphism ([11], III, 2.2)

$$d : R_K(G) \longrightarrow R_k(G) :$$

for V a representation of G over K , and E a G -invariant lattice,

$$d([V]) = [E \otimes_A k].$$

Proposition 1.8. Assume that K is sufficiently large. The kernel of the decomposition homomorphism d is then generated by the elements

$$\text{Ind}_H^G(\chi') - \text{Ind}_H^G(\chi'')$$

where H is an elementary subgroup and $\chi', \chi'' \in \text{Hom}(H, K^*)$ are congruent modulo $\mathfrak{m}$.

Since the ring homomorphism d commutes with induction, the argument of 1.5 reduces us to the case where G is elementary, hence the product of a p -group P and a group H of order prime to p . Since K and k are sufficiently large, one has

$$R(G) = R(P) \otimes R(H),$$

and similarly over k . For a group of order prime to p , d is bijective. Applying Brauer's theorem to H , it is enough to prove 1.8 for a p -group. This case is trivial, since every character $\chi \in \text{Hom}(P, K^*)$ of a p -group is congruent to 1 modulo $\mathfrak{m}$.

1.9. For the rest of this subsection, assume that $K = \mathbb{C}$, consider only finite groups, and abbreviate $R_{\mathbb{C}}(G)$ to $R(G)$. For G commutative, write G^* for the group of characters of G .

Let $R^+(G)$ be the free $\mathbb{Z}$ -module with basis the set of G -conjugacy classes of pairs (H, χ) , where H is a subgroup of G and χ a character of H . Define a multiplication in $R^+(G)$ by

$$(A, \alpha)(B, \beta) = \sum_{(x, y) \in G \backslash (G/A \times G/B)} (A^x \cap B^y, (\alpha^x|_{A^x \cap B^y})(\beta^y|_{A^x \cap B^y})), \quad (1.9.1)$$

where (x, y) runs through representatives of the orbits of G in $G/A \times G/B$, $A^x = xAx^{-1}$, and α^x is the character $a \mapsto \alpha(x^{-1}ax)$ of A^x . With this multiplication, $R^+(G)$ is a commutative ring with unit $(G, 1)$, and one defines a ring homomorphism

$$b : R^+(G) \longrightarrow R(G), \quad (H, \chi) \longmapsto \text{Ind}_H^G(\chi). \quad (1.9.2)$$

This homomorphism is surjective by Brauer's theorem. For χ a character of G , multiplication by (G, χ) is denoted $\cdot \chi$ and is called twisting by χ :

$$(H, \varphi) \cdot \chi = (H, \varphi \cdot \chi|_H). \quad (1.9.3)$$

For $G' \subset G$, define induction

$$\text{Ind}_{G'}^G : R^+(G') \longrightarrow R^+(G), \quad (H, \chi) \longmapsto (H, \chi). \quad (1.9.4)$$

For a morphism $u : G \rightarrow G'$, define restriction by

$$\text{Res}_u(H, \chi) = \sum_{s \in u(G) \backslash G' / H} (u^{-1}(sHs^{-1}), \chi^s \circ u), \quad (1.9.5)$$

where s runs over a set of representatives of the double classes. There are commutative diagrams

$$\begin{array}{ccc} R^+(G') & \xrightarrow{\text{Ind}} & R^+(G) \\ b \downarrow & & \downarrow b \\ R(G') & \xrightarrow{\text{Ind}} & R(G) \end{array} \quad \begin{array}{ccc} R^+(G') & \xrightarrow{\text{Res}} & R^+(G) \\ b \downarrow & & \downarrow b \\ R(G') & \xrightarrow{\text{Res}} & R(G), \end{array}$$

for the second, see [11], p. 75. For $R \in R^+(G)$ one has

$$R \cdot (H, \chi) = \text{Ind}_H^G (\text{Res}_H^G(R) \cdot \chi). \quad (1.9.6)$$

When u is surjective, one speaks of inflation.

Notation 1.9.8. For every additive function of representations F , write the same symbol for the function it defines on $R(G)$, and for $F \circ b$. Thus

$$\dim(v) = \dim(b(v)), \quad \dots$$

Variants 1.10. (i) Let $R_0^+(G)$ be the subgroup of $R^+(G)$ generated by the $(H, \chi) - (H, 1)$. As (H, χ) runs through the canonical basis of $R^+(G)$, except for the $(H, 1)$, these elements form a basis of $R_0^+(G)$. One checks that $R_0^+(G)$ is an ideal of $R^+(G)$. Let $R^0(G) \subset R(G)$ be the set of virtual representations of dimension 0 of G . By 1.5, the natural map, induced by (1.9.2), from $R_0^+(G)$ to $R^0(G)$ is surjective.

(ii) For a topological group G , finite or not, $R^+(G)$ and $R_0^+(G)$ denote the analogous groups obtained by restricting to closed subgroups H of finite index and continuous quasi-characters. The functorialities of 1.9 generalize mutatis mutandis.

The rest of this subsection will not be used below. We give results drawn from Langlands [9].

Lemma 1.11. Suppose that $G = H \cdot C$, where C is commutative and normal, and let χ be a character of H . For every $\mu \in C^*$, let $G_\mu \subset G$ be the stabilizer of μ under conjugation. If

$$\chi|_{H \cap C} = \mu|_{H \cap C},$$

write $\{\chi, \mu\}$ for the character of $G_\mu = (H \cap G_\mu) \cdot C$ which extends $\chi|_{H \cap G_\mu}$ and μ . Then

$$\text{Ind}_H^G(\chi) = \sum_{\substack{\mu \in C^*/G \\ \chi|_{H \cap C} = \mu|_{H \cap C}}} \text{Ind}_{G_\mu}^G \{\chi, \mu\}.$$

The verification is left to the reader.

1.12. An element R of $\text{Ker}(R^+(G) \rightarrow R(G))$ will be said to be of type I, II, or III if there exists a quotient A of a subgroup B of G such that R is obtained from an element S of one of the following respective types in $\text{Ker}(R^+(A) \rightarrow R(A))$, by inflation from A to B , twisting by a character of B , and induction from B to G . Let ℓ be a prime number.

I. $A \simeq \mathbb{Z}/\ell\mathbb{Z}$, and

$$S = (e, 1) - \sum_{\chi \in A^*} (A, \chi).$$

II. A is a central extension of $(\mathbb{Z}/\ell\mathbb{Z})^2$ by an abelian group Z . Let H_i , $i = 1, 2$, be the inverse images in A of the first and second factors $\mathbb{Z}/\ell\mathbb{Z}$, and let χ_i be a character of H_i . Assume that $\chi_1|_Z = \chi_2|_Z$, and that this character of Z is nontrivial on a commutator.

$$\begin{array}{ccc} Z & \hookrightarrow & H_1 \\ \downarrow & & \downarrow \\ H_2 & \hookrightarrow & A \end{array}$$

The relation S is

$$S = (H_1, \chi_1) - (H_2, \chi_2).$$

III. A is a semidirect product $H \cdot C$, χ is a character of H , and the normal subgroup C is minimal among the nontrivial commutative normal subgroups of A . For every character μ of C , A_μ is the stabilizer of μ , and $\{\chi, \mu\}$ is the character of A_μ extending χ and μ . The relation S expresses

$$\text{Ind}_H^A(\chi) = \sum_{\mu \in C^*/A} \text{Ind}_{A_\mu}^A(\{\chi, \mu\}).$$

where μ runs over a set of representatives for the orbits of A in C^* .

These relations are special cases of 1.11: for I, $H = \{e\}$ and $C = A$; for II, $H = H_1$ and $C = H_2$; for III, H and C are as indicated.

Every relation obtained by induction, inflation, or twisting from a relation of type I, respectively II or III, is again of the same type.

Theorem 1.13 (Langlands [9], §18). If G is nilpotent, the kernel of

$$b : R^+(G) \longrightarrow R(G)$$

is generated, as a $\mathbb{Z}$ -module, by the relations of type I and II.

Lemma 1.13.1. If G is commutative, $\text{Ker}(b)$ is generated by the relations of type I.

One has to show that, for every subgroup H of G and every $\chi \in H^*$, the relation

$$(H, \chi) - \sum_{\psi \in (G/H)^*} (G, \tilde{\chi}\psi)$$

is a consequence of relations of type I. These are the following: for $H' \subset H'' \subset G$, with $[H'' : H']$ prime, and χ a character of H' , always induced by a character of G ,

$$\text{Ind}_{H'}^G(\chi) - \sum_{\chi''|_{H'}=\chi} \text{Ind}_{H''}^G(\chi'') = 0.$$

The general relation follows by taking a Jordan–Hölder series of G/H .

Lemma 1.13.2. Let C be a commutative normal subgroup of G , let T be a set of representatives of C^*/G , let $\mu \in T$, and let G_μ be the stabilizer of μ . Let (H_i) be a family of subgroups of G containing C , and let χ_i be a character of H_i , with $\chi_i|_C \in T$. If the relation

$$\sum_i n_i \text{Ind}_{H_i}^G(\chi_i) = 0$$

holds in $R(G)$, then

$$\sum_{\chi_i|_C=\mu} n_i \text{Ind}_{H_i}^{G_\mu}(\chi_i) = 0$$

holds in $R(G_\mu)$. The verification is left to the reader.

Let Z be the center of G , and prove 1.13 by induction on the order of G/Z . By 1.13.1, we may suppose G noncommutative. Let C be a commutative normal subgroup containing Z , with $[C : Z] = \ell$ prime, and with central image in G/Z .

Consider a relation

$$\sum_i n_i \text{Ind}_{H_i}^G(\chi_i) = 0. \quad (1)$$

It is a consequence of relations induced by

$$\sum_{\chi_i|_{H_i \cap Z} = \chi} \text{Ind}_{H_i}^G(\chi_i) = \text{Ind}_{H_i Z}^G(\chi)$$

and of a relation (2) analogous to (1), with $H_i \supset Z$. The displayed relations are linear combinations of relations of type I; this follows from 1.13.1 applied to $H_i Z / \text{Ker}(\chi_i)$.

The relation (2) is a consequence of relations induced from 1.11 for HC, H, C, χ , with $H \supset Z$,

$$\text{Ind}_H^{HC}(\chi) = \sum_{\substack{\mu \in C^*/H \\ \chi|_{H \cap C} = \mu|_{H \cap C}}} \text{Ind}_{G_\mu}^{HC}\{\chi, \mu\}, \quad (b)$$

and of a relation (3) analogous to (1), this time with $H_i \supset C$. Apply 1.13.2 to (3), to express it as a sum of relations induced by relations

$$\sum_i n_i \text{Ind}_{H_i}^{G_\mu}(\chi_i) = 0.$$

The image of C in $G/\text{Ker}(\mu)$ is central, and the induction hypothesis therefore applies to G_μ ; the resulting relations are of the desired type.

It remains to prove that the relations (b) are of the desired type. If $HC \neq G$, this follows from the induction hypothesis applied to HC . If $HC = G$, then H is normal, because it contains Z and C is central modulo Z . Let A be the intersection of the kernels of the conjugates of χ . The relation (b) is the inflation of an analogous relation for G/A . If the induction hypothesis applies to G/A , we are done. Otherwise we are in the same situation as before, with $A = \{e\}$, and we conclude by the following lemma.

Lemma 1.13.3. With the preceding notation, if $A = \{e\}$, the relation (b) is of type II.

The group H is commutative, since characters separate its elements. The group Z is cyclic, since one character separates its elements. For $c \in C$ and $h \in H$, put

$$u(c, h) = chc^{-1}h^{-1}.$$

This is an element of Z , since C is central modulo Z . It depends biadditively on c and h , and defines

$$u : C/Z \otimes H/Z \longrightarrow Z.$$

Let c generate C/Z . Since Z is the center and $G \neq Z$, the map $u(c, \cdot) : H/Z \rightarrow Z$ is injective, with nontrivial cyclic image killed by ℓ . Hence $|C/Z| = |H/Z| = \ell$, and G is a central extension of $C/Z \times H/Z$ by Z . The character χ is distinct from at least one of its conjugates, otherwise H would be central; it is therefore nontrivial on a commutator, and we are in situation II.

Theorem 1.14 (Langlands [9]). If G is solvable, the kernel of

$$b : R^+(G) \longrightarrow R(G)$$

is generated by the relations of type I, II, and III.

Lemma 1.14.1. If G is solvable, the $\mathbb{Z}$ -submodule $N(G)$ of $R^+(G)$ generated by the relations of type I, II, and III is an ideal.

We prove 1.14 and 1.14.1 by simultaneous induction on the order of G .

(A) Theorem 1.14 for the proper subgroups of G implies 1.14.1 for G . Let $R \in R^+(G)$ be of type I, II, or III, let H be a subgroup, and let χ be a character of H . We prove that

$$R \cdot (H, \chi) \in N(G).$$

If $H = G$, this follows from the stability of $N(G)$ under twisting by a character of G . If $H \neq G$, then by (1.9.6)

$$R \cdot (H, \chi) = \text{Ind}_H^G(\text{Res}_H^G(R) \cdot \chi).$$

By the hypothesis, $\text{Res}_H^G(R) \in N(H)$, and twisting followed by induction gives an element of $N(G)$.

(B) Lemma 1.14.1 for G , together with Theorem 1.14 for groups of smaller order, implies 1.14 for G . Let Z be the center of G . By 1.13, we may and shall suppose that G is not nilpotent. There are two cases.

(B1) $Z \neq \{e\}$. By Brauer's theorem in G/Z , there exist nilpotent subgroups $H_i \supset Z$ of G , and characters χ_i of H_i/Z , such that

$$\sum_i n_i \text{Ind}_{H_i/Z}^{G/Z}(\chi_i) = 1_{G/Z}.$$

Since this relation lifts to a relation S in $R^+(G)$, and since the induction hypothesis applies to G/Z , one has $S - (G, 1) \in N(G)$. For $R \in \text{Ker}(b)$, one has $R \cdot S \in N(G)$ by 1.14.1, while (1.9.6) reduces the terms to restrictions to the H_i , where the induction hypothesis applies. This proves the assertion.

(B2) $Z = \{e\}$. Let C be a nontrivial minimal commutative normal subgroup and, for $\mu \in C^*$, let G_μ be its stabilizer. Proceed as in the proof of 1.13. One finds that every relation $R \in \text{Ker}(b)$ is a linear combination of:

- (a) relations induced from 1.11 for HC, H, C, χ , with $H \subset G$, χ a character of H , and $H \not\supset C$;
- (b) relations induced from identities in G_μ .

Since $Z = \{e\}$, one has $|G_\mu / \text{Ker}(\mu)| < |G|$, and the induction hypothesis applies to the latter relations. Consider a relation of type (a). If $HC \neq G$, apply the induction hypothesis. If $HC = G$, then $H \cap C$ is normal, since it is normal in both H and C . By minimality of C , $H \cap C = \{e\}$, and the relation is of type III.

Remark 1.15. It is probably possible to extract from [9] a description of a system of generators of

$$\text{Ker}(R_0^+(G) \longrightarrow R^0(G))$$

for suitable groups G . I have not had the courage to try.

2. Weil groups

2.1. Roots of unity. Let K be a separably closed field of characteristic exponent p . For every integer n , write $\mu_n(1)(K)$, or simply $\mu_n(1)$, for the group of n -th roots of unity in K . For $n \mid m$, there is a transition map

$$\mu_m(1) \longrightarrow \mu_n(1), \quad x \longmapsto x^{m/n};$$

put

$$\widehat{\mathbb{Z}}(1) = \varprojlim_n \mu_n(1).$$

For a prime number ℓ , similarly put

$$\mathbb{Z}_\ell(1) = \varprojlim_r \mu_{\ell^r}(1).$$

The evident maps $\widehat{\mathbb{Z}}(1) \rightarrow \mathbb{Z}_\ell(1)$ induce an isomorphism

$$\widehat{\mathbb{Z}}(1) \xrightarrow{\sim} \prod_{\ell \neq p} \mathbb{Z}_\ell(1).$$

For every $\widehat{\mathbb{Z}}$ -module V , put $V(1) = V \otimes_{\widehat{\mathbb{Z}}} \widehat{\mathbb{Z}}(1)$. For a $\mathbb{Z}_\ell$ -module V , one has

$$V(1) \simeq V \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1).$$

For $K = \mathbb{C}$, the group $\mu_n(1)$ is canonically isomorphic to the group of roots of unity in $\mathbb{C}$. For $x \in \mu_n(1)$ and $\xi \in \mathbb{Z}/n\mathbb{Z}$, the image of $\xi \otimes x$ in $\mathbb{Q}/\mathbb{Z}(1)$ is ξx , giving the usual identifications.

For $\ell \neq p$, $\mathbb{Z}_\ell(1)$ is a free rank-one $\mathbb{Z}_\ell$ -module. For $i \in \mathbb{Z}$, write $\mathbb{Z}_\ell(i)$ for its i -th tensor power, and for $i < 0$ for the dual of $\mathbb{Z}_\ell(-i)$. For every $\mathbb{Z}_\ell$ -module V , put again $V(i) = V \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(i)$.

2.2. Notation. Let K be a local field. We write $\| \cdot \|$ for the normalized absolute value of K . If dx is a Haar measure, then

$$\int f(ax) dx = \|a\|^{-1} \int f(x) dx,$$

i.e. $d(ax) = \|a\| dx$.

For $s \in \mathbb{C}$, write ω_s for the quasi-character

$$x \mapsto \|x\|^s$$

from K^* to $\mathbb{C}^*$.

If K is nonarchimedean, i.e. $K \neq \mathbb{R}, \mathbb{C}$, write $\mathcal{O}$ for the valuation ring of the valuation v of K , π for a uniformizer, k for the residue field $\mathcal{O}/(\pi)$, p for the characteristic of k , and put

$$d = [k : \mathbb{F}_p], \quad q = p^d = \#k.$$

Thus

$$\|x\| = q^{-v(x)}.$$

2.2.2. Suppose K is nonarchimedean. Let $\overline{K}$ be a separable closure of K , write $\overline{\mathcal{O}}$ for the integral closure of $\mathcal{O}$ in $\overline{K}$, let $\overline{k}$ be the residue field of $\overline{\mathcal{O}}$, an algebraic closure of k , and let v be the valuation of $\overline{K}$, with values in $\mathbb{Q}$, extending that of K .

There is a three-step filtration of the Galois group

$$\text{Gal}(\overline{K}/K) \supset I \supset P$$

whose successive quotients are

$$\begin{aligned} \text{Gal}(\overline{K}/K)/I &\xrightarrow{\sim} \text{Gal}(\overline{k}/k) \xrightarrow{\sim} \widehat{\mathbb{Z}}, \\ I/P &\xrightarrow{\sim} \widehat{\mathbb{Z}}(1)(\overline{k}), \\ P &: \text{ a pro-}p\text{-group.} \end{aligned}$$

It is obtained as follows; for proofs we refer to [10].

- a) By transport of structure, $\text{Gal}(\overline{K}/K)$ acts on $\overline{\mathcal{O}}$ and on $\overline{k}$. The inertia group I is the kernel of the reduction map

$$v' : \text{Gal}(\overline{K}/K) \longrightarrow \text{Gal}(\overline{k}/k).$$

The quotient $\text{Gal}(\overline{k}/k)$ is canonically isomorphic to $\widehat{\mathbb{Z}}$, with topological generator the Frobenius substitution

$$\varphi : x \mapsto x^q.$$

- b) The wild inertia group P is the unique pro- p -Sylow subgroup of I . The equivariant morphism

$$t : I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}),$$

with kernel P , is characterized by the congruence, modulo the maximal ideal of $\overline{\mathcal{O}}$,

$$\frac{\sigma(x)}{x} = t(\sigma) v(x) \quad (\sigma \in I, x \in \overline{K}).$$

For a prime $\ell \neq p$, write t_ℓ for the composite

$$I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}) \longrightarrow \mathbb{Z}_\ell(1)(\overline{k}).$$

2.2.3. Let k be a finite field with q elements, and let $\bar{k}$ be its algebraic closure. Let $W(\bar{k}/k)$ be the subgroup of $\text{Gal}(\bar{k}/k) \simeq \widehat{\mathbb{Z}}$ generated by

$$\varphi : x \mapsto x^q.$$

There is a canonical isomorphism

$$W(\bar{k}/k) \simeq \mathbb{Z}$$

with generator φ . The geometric Frobenius $F \in W(\bar{k}/k)$ is, by definition, φ^{-1} .

2.2.4. Let $K, \bar{K}$ be as in 2.2.2. The Weil group $W(\bar{K}/K)$, or absolute Weil group of K , is the subgroup of $\text{Gal}(\bar{K}/K)$ consisting of the σ such that $v'(\sigma)$ is an integral power of Frobenius φ . It is endowed with the topology inducing the natural topology on I , and for which I is open. The group $\text{Gal}(\bar{K}/K)$ is the completion of $W(\bar{K}/K)$ for the topology of open subgroups of finite index; if L is a finite extension of K in $\bar{K}$, the corresponding subgroup of $W(\bar{K}/K)$ is identified with $W(\bar{K}/L)$. We shall sometimes call geometric Frobenius elements those elements of $\text{Gal}(\bar{K}/K)$ or $W(\bar{K}/K)$ whose image is F .

2.2.5. Suppose K is archimedean, with algebraic closure $\bar{K}$. Define $W(\bar{K}/K)$ as the following extension of $\text{Gal}(\bar{K}/K)$ by $\bar{K}^*$:

- 1) $K \simeq \mathbb{R}$: $W(\bar{K}/K)$ is generated by $\bar{K}^*$ and an element F , with

$$F^2 = -1, \quad FzF^{-1} = \bar{z} \quad (z \in \bar{K}^*).$$

- 2) $K \simeq \mathbb{C}$: $W(\bar{K}/K) = \bar{K}^*$.

2.3. Local class field theory provides an isomorphism between $W(\bar{K}/K)^{\text{ab}}$ and K^* .

Assume K nonarchimedean. For a reason explained in 3.6, we normalize this isomorphism, i.e. choose it or its inverse, so that geometric Frobenius elements correspond to uniformizers:

$$\begin{array}{ccccccc} P & \hookrightarrow & I & \hookrightarrow & W(\bar{K}/K) & \longrightarrow & W(\bar{k}/k) \simeq \mathbb{Z} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow -1 \\ 1 + (\pi) & \hookrightarrow & \mathcal{O}^* & \hookrightarrow & K^* & \xrightarrow{v} & \mathbb{Z}. \end{array}$$

A representation of $W(\bar{K}/K)$ is unramified, respectively tamely ramified, if it is trivial on I , respectively on P . Similarly, a quasi-character χ of K^* is unramified, respectively tamely ramified, if it is trivial on $\mathcal{O}^*$, respectively on $1 + (\pi)$. For ℓ prime to p , the action of $\text{Gal}(\bar{K}/K)$ on $\mathbb{Z}_\ell(1)$ is unramified, described by the quasi-character

$$\omega_1 = \|x\|$$

of K^* .

For K complex, the isomorphism of local class field theory is the evident one on 2.2.5; for K real, it is the one which makes 2.3.1 and 2.3.2 below true.

Let $L \subset \bar{K}$ be a finite extension of K , so that $W(\bar{K}/L) \subset W(\bar{K}/K)$. The following diagrams, where N is the norm and t the transfer, are commutative ([10], XIII, §4):

$$\begin{array}{ccccc} W(\bar{K}/L) & \longrightarrow & W(\bar{K}/L)^{\text{ab}} & \xrightarrow{\sim} & L^* \\ \downarrow & & \downarrow & & \downarrow N_{L/K} \\ W(\bar{K}/K) & \longrightarrow & W(\bar{K}/K)^{\text{ab}} & \xrightarrow{\sim} & K^*, \end{array} \tag{2.3.1}$$

$$\begin{array}{ccc} W(\bar{K}/K)^{\text{ab}} & \xrightarrow{\sim} & K^* \\ t \downarrow & & \downarrow \\ W(\bar{K}/L)^{\text{ab}} & \xrightarrow{\sim} & L^*. \end{array} \tag{2.3.2}$$

2.4. Let K be a function field in one variable over $\mathbb{F}_p$, and let k be its field of constants, the algebraic closure of $\mathbb{F}_p$ in K . Let $\overline{K}$ be a separable closure of K , let $\overline{k}$ be the algebraic closure of k in $\overline{K}$, and let $\text{Gal}^0(\overline{K}/K)$ be the kernel of the restriction map from $\text{Gal}(\overline{K}/K)$ to $\text{Gal}(\overline{k}/k)$. The absolute global Weil group $W(\overline{K}/K)$ is, by analogy with 2.2.4, defined by the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Gal}^0(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k) = \mathbb{Z} \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Gal}^0(\overline{K}/K) & \longrightarrow & \text{Gal}(\overline{K}/K) & \longrightarrow & \text{Gal}(\overline{k}/k) = \widehat{\mathbb{Z}} \longrightarrow 0. \end{array}$$

Let v be a place of K , and suppose a place of $\overline{K}$ above v has been chosen. There is then a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I_v & \longrightarrow & W(\overline{K}_v/K_v) & \longrightarrow & W(\overline{k}_v/k_v) = \mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow [k_v:k] \\ 0 & \longrightarrow & \text{Gal}^0(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k) = \mathbb{Z} \longrightarrow 0. \end{array}$$

Global class field theory provides an isomorphism

$$W(\overline{K}/K)^{\text{ab}} \xrightarrow{\sim} \mathbb{A}_K^*/K^*$$

where $\mathbb{A}_K^*/K^*$ is the idèle-class group, making the diagrams

$$\begin{array}{ccc} W(\overline{K}_v/K_v)^{\text{ab}} & \longrightarrow & W(\overline{K}/K)^{\text{ab}} \\ \downarrow & & \downarrow \\ K_v^* & \longrightarrow & \mathbb{A}_K^*/K^* \end{array}$$

commutative.

2.5. For the definition of global Weil groups in the case of number fields, we refer to Weil [19] or to [2], XIV. We shall use them only incidentally.

3. Reminders on L -functions

3.1. Let K be a local field and use again the notation of 2.2.1. We denote by dx a Haar measure on K , by d^*x a Haar measure on K^* , and by ψ a nontrivial additive character of K .

3.2. For a quasi-character, i.e. a continuous homomorphism,

$$\chi : K^* \longrightarrow \mathbb{C}^*,$$

define $L(\chi) \in \mathbb{C} \cup \{\infty\}$ as follows.

$$K \simeq \mathbb{R}. \tag{3.2.1}$$

Put

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2),$$

and, for x the embedding of K in $\mathbb{C}$ and $N = 0$ or 1 ,

$$L(x^{-N} \omega_s) = \Gamma_{\mathbb{R}}(s).$$

$$K \simeq \mathbb{C}. \tag{3.2.2}$$

Put

$$\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s),$$

and, for z an embedding of K in $\mathbb{C}$ and $N \geq 0$,

$$L(z^{-N}\omega_s) = \Gamma_{\mathbb{C}}(s).$$

K nonarchimedean.

(3.2.3)

Put

$$L(\chi) = 1 \quad (\chi \text{ ramified}), \quad L(\chi) = \frac{1}{1 - \chi(\pi)} \quad (\chi \text{ unramified}).$$

3.3. Given ψ and dx , put, for f a C^∞ function of compact support on K ,

$$\widehat{f}(y) = \int_K f(x)\psi(xy) dx.$$

Define $\varepsilon(\chi, \psi, dx) \in \mathbb{C}^*$ by Tate's local functional equation, independent of f ,

$$\frac{\int_{K^*} \widehat{f}(x) \omega_1 \chi^{-1}(x) d^*x}{L(\omega_1 \chi^{-1})} = \varepsilon(\chi, \psi, dx) \frac{\int_{K^*} f(x) \chi(x) d^*x}{L(\chi)}. \quad (3.3.1)$$

The same Haar measure d^*x is used on both sides. Both sides are defined by analytic continuation in χ in the families $\chi\omega_s$, $s \in \mathbb{C}$. In these families, ε is of the form Ae^{Bs} .

One has

$$\varepsilon(\chi, \psi, a dx) = a \varepsilon(\chi, \psi, dx), \quad (3.3.2)$$

$$\varepsilon(\chi, \psi(ax), dx) = \chi(a) \|a\|^{-1} \varepsilon(\chi, \psi, dx). \quad (3.3.3)$$

3.4. For K nonarchimedean, put:

$$n(\psi) = \text{the largest integer } n \text{ such that } \psi|_{\pi^{-n}\mathcal{O}} = 1;$$

$$a(\chi) = \begin{cases} 0, & \chi \text{ unramified,} \\ \text{the least } m \text{ such that } \chi|_{1+(\pi)^m} = 1, & \chi \text{ ramified;} \end{cases}$$

$$sw(\chi) = \begin{cases} 0, & \chi \text{ unramified or tamely ramified,} \\ a(\chi) - 1, & \chi \text{ ramified;} \end{cases}$$

and let γ be an element of K^* of valuation

$$v(\gamma) = n(\psi) + sw(\chi) + 1.$$

The local constant ε is given by (3.3.2), (3.3.3), and the following formulas.

$$K \simeq \mathbb{R}. \quad (3.4.1)$$

Let x be the embedding of K in $\mathbb{C}$, and $N = 0$ or 1 . For $\psi = \exp(2\pi i x)$ and dx the Lebesgue measure,

$$\varepsilon(x^{-N}\omega_s, \psi, dx) = i^N.$$

$$K \simeq \mathbb{C}. \quad (3.4.2)$$

Let z be an embedding of K in $\mathbb{C}$, and $N \geq 0$. For

$$\psi = \exp(2\pi i \operatorname{Tr}_{\mathbb{C}/\mathbb{R}}(z))$$

and

$$dx = |dz \wedge d\bar{z}| = 2 da db \quad (z = a + bi),$$

one has

$$\varepsilon(z^{-N}\omega_s, \psi, dx) = i^N.$$

3.4.3. K nonarchimedean. If χ is unramified,

$$\int_{\mathcal{O}} dx = 1, \quad n(\psi) = 0,$$

then

$$\varepsilon(\chi, \psi, dx) = 1. \quad (3.4.3.1)$$

For χ ramified, one has

$$\varepsilon(\chi, \psi, dx) = \int_{K^*} \chi^{-1}(x) \psi(x) dx = \sum_n \int_{v(x)=n} \chi^{-1}(x) \psi(x) dx = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x) \psi(x) dx. \quad (3.4.3.2)$$

We shall see that it is natural to unify these two formulas as follows. If $\varepsilon_0(\chi, \psi, dx)$ is defined by

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx) \quad (\chi \text{ ramified}),$$

and

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx) \chi(\pi)^{-1} \quad (\chi \text{ unramified}), \quad (3.4.3.3)$$

then

$$\varepsilon_0(\chi, \psi, dx) = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x) \psi(x) dx. \quad (3.4.3.4)$$

From (3.4.3.3) and (3.4.3.4), one deduces that, for ω unramified,

$$\varepsilon_0(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+sw(\chi)+1} \varepsilon_0(\chi, \psi, dx), \quad (3.4.3.5)$$

and

$$\varepsilon(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+a(\chi)} \varepsilon(\chi, \psi, dx).$$

3.5. Suppose K nonarchimedean, and resume the notation of 2.2.1. Let V be a finite-dimensional vector space over a field E of characteristic 0, or over $\mathbb{C}$. The representations

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

will always be assumed continuous: an open subgroup of I acts trivially. For V the space of a representation and V^I the subspace of inertia invariants, put

$$Z(V; t) = \det(1 - Ft^d, V^I)^{-1}, \quad (3.5.1)$$

$$L(V) = Z(V; 1) \quad (E \in E^* \cup \{\infty\}). \quad (3.5.2)$$

Let

$$\omega^t : W(\overline{K}/K) \longrightarrow E((t))^*$$

be the unramified character such that $\omega^t(F) = t^d$. In an evident sense, $\omega_s = \omega^{p^{-s}}$. One has

$$Z(V, t) = L(V \otimes \omega^t). \quad (3.5.3)$$

3.6. It is so that (3.2.3) and (3.5.2) are compatible that we normalized the local class field theory isomorphism so that geometric Frobenius elements correspond to uniformizers. Both (3.2.3) and (3.5.2) are very natural: 3.2.3 is forced from the analytic point of view (cf. 3.3.1), and 3.5.2 follows the tradition of geometers; it is geometric Frobenius which tends to act on the cohomology of algebraic varieties with algebraic integers as eigenvalues.

3.7. The irreducible continuous complex representations of $W(\overline{K}/K)$ are, for K complex, the quasi-characters. For K real, besides the quasi-characters of K^* , they are the representations induced by quasi-characters χ of $\overline{K}^*$ such that $\chi \neq \chi \circ \sigma$, where σ is complex conjugation of $\overline{K}$.

For any complex representation V , expressed in the Grothendieck group of representations of $W(\overline{K}/K)$ as a sum of simple representations, define $L(V)$ as follows:

$$K \simeq \mathbb{R}, \quad V = \sum_i [\chi_i] + \sum_j \text{Ind}_{\overline{K}^*}^W(\chi_j), \quad (3.7.1)$$

$$L(V) = \prod_i L_K(\chi_i) \prod_j L_{\overline{K}}(\chi_j).$$

$$K \simeq \mathbb{C}, \quad V = \sum_i [\chi_i], \quad L(V) = \prod_i L(\chi_i). \quad (3.7.2)$$

Proposition 3.8. Let K be a local field and $\overline{K}$ an algebraic closure of K .

(i) For every exact sequence

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0$$

of complex representations of $W(\overline{K}/K)$, one has

$$L(V) = L(V')L(V''). \quad (3.8.1)$$

(ii) Let L be a finite separable extension of K in $\overline{K}$, and let V_L be a complex representation of $W(\overline{K}/L)$. Let V_K be the representation induced to $W(\overline{K}/K)$. Then

$$L(V_K) = L(V_L). \quad (3.8.2)$$

Assertion (i) is trivial. In the archimedean case, assertion (ii) follows from the duplication formula

$$\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1),$$

corresponding to

$$\text{Ind}([\omega_s]) = [\omega_s] + [\chi^{-1}\omega_{s+1}].$$

We prove (ii) for K nonarchimedean. Let k and ℓ be the residue fields of K and L . Then

$$V_K^{I_K} = \left(\text{Ind}_{W(\overline{K}/L)}^{W(\overline{K}/K)} V_L \right)^{I_K} \simeq \text{Ind}_{W(\overline{\ell}/\ell)}^{W(\overline{k}/k)} (V_L^{I_L})$$

as $W(\overline{k}/k)$ -representations, both sides being identified with

$$\text{Hom}_{W(\overline{K}/L)}(W(\overline{k}/k), V_L),$$

the covariant functions from $W(\overline{k}/k)$ to V_L , $W(\overline{K}/L)$ acting on $W(\overline{k}/k)$ by translation; cf. the commutative diagram

$$\begin{array}{ccc} W(\overline{K}/L) & \longrightarrow & W(\overline{\ell}/\ell) \\ \downarrow & & \downarrow \\ W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k). \end{array}$$

This reduces (3.8.2) to the following well-known lemma.

Lemma 3.9. Let F be an endomorphism of V , let n be an integer, and let F_n be the endomorphism of

$$V_n = V \otimes \mathbb{C}^n$$

such that, for a basis $e_0, \dots, e_{n-1}$ of $\mathbb{C}^n$,

$$F_n(v \otimes e_i) = v \otimes e_{i+1} \quad (0 \leq i < n-1), \quad F_n(v \otimes e_{n-1}) = F(v) \otimes e_0.$$

Then

$$\det(1 - F_n t) = \det(1 - F t^n).$$

By the principle of continuation of algebraic identities, one may suppose F diagonal in a suitable basis of V . One is immediately reduced to the case $\dim(V) = 1$, where F is multiplication by a number a^n , $a \neq 0$. In the basis $f_i = a^{-i}(v \otimes e_i)$, one has $F_n(f_i) = a f_{i+1}$, for $i \in \mathbb{Z}/n\mathbb{Z}$. The eigenvectors of F_n are the

$$\sum_i \chi(i) f_i$$

where χ runs through the characters of $\mathbb{Z}/n\mathbb{Z}$, and the determinant identity follows.

3.10. Let K be a global field, an $\mathfrak{A}$ -field in the sense of [15]. For every place v of K , write K_v for the local field which is the completion of K at v . The objects defined above for local fields will be denoted, for K_v , with a subscript v . Let $\mathbb{A}$ be the adele ring of K , let $\| \cdot \|$ be the absolute value satisfying

$$d(ax) = \|a\| dx$$

for a Haar measure dx on $\mathbb{A}$, and let ω_s be the quasi-character $\|x\|^s$ of $\mathbb{A}^*$. If x_v is the component of $x \in \mathbb{A}$ in K_v , then

$$\|x\| = \prod_v \|x_v\|_v.$$

Let dx denote the unique Haar measure on $\mathbb{A}$ such that

$$\int_{\mathbb{A}/K} dx = 1$$

(the Tamagawa measure), let d^*x be a Haar measure on $\mathbb{A}^*$, and let ψ be a character trivial on $\mathbb{A}/K$, with local components ψ_v . For f a C^∞ function with compact support on $\mathbb{A}$, put

$$\hat{f}(y) = \int_{\mathbb{A}} f(x) \psi(xy) dx.$$

3.11. Tate's global functional equation [14] says, for χ a quasi-character of $\mathbb{A}^*/K^*$,

$$\int_{\mathbb{A}^*} \widehat{f}(x) \chi'(x) d^*x = \int_{\mathbb{A}^*} f(x) \chi(x) d^*x, \quad \chi' = \omega_1 \chi^{-1}, \quad (3.11.1)$$

the integrals being defined by analytic continuation in the families $\chi\omega_s$. Put

$$L(\chi) = \prod_v L(\chi_v) \quad (3.11.2)$$

by analytic continuation, and, for $dx = \otimes_v dx_v$,

$$\varepsilon(\chi) = \prod_v \varepsilon(\chi_v, \psi_v, dx_v). \quad (3.11.3)$$

The product (3.11.3) is independent of the choice of ψ and of the decomposition of dx , as follows from (3.3.2) and (3.3.3). If, for almost all v ,

$$\int_{\mathcal{O}_v} dx_v = 1,$$

then almost all its factors are equal to 1. Formulas (3.3.1) and (3.11.1) give the global functional equation of Hecke L -functions:

$$\varepsilon(\chi) L(\chi') = L(\chi). \quad (3.11.4)$$

3.12. The results stated below are proved in [1] and [19]. Let K be a global field, $\overline{K}$ an algebraic closure of K , and V a complex representation of $\text{Gal}(\overline{K}/K)$, always assumed continuous: the action of $\text{Gal}(\overline{K}/K)$ factors through a finite Galois group $\text{Gal}(K'/K)$. For every place v of K , choose a place of $\overline{K}$ above it, and let V_v be the representation obtained from V by restriction to the subgroup

$$W(\overline{K}_v/K_v)$$

of the decomposition group $\text{Gal}(\overline{K}_v/K_v)$.

(A) For χ a quasi-character of $\mathbb{A}^*/K^*$, with local components χ_v , the infinite product

$$L(V, \chi) = \prod_v L(V_v \otimes \chi_v)$$

can be defined by analytic continuation in χ in the families $\chi\omega_s$. It converges for $\Re(s)$ large and extends to a meromorphic function of s .

(B) Let V^* denote the dual of V . One has

$$L(V, \chi) = \varepsilon(V, \chi) L(V^*, \chi'), \quad \chi' = \omega_1 \chi^{-1},$$

with $\varepsilon(V, \chi) \in \mathbb{C}^*$, of the form Ae^{Bs} as χ varies in the families $\chi\omega_s$.

(C) Let L be a finite extension of K in $\overline{K}$, let $N_{L/K}$ be the norm, let V_L be a representation of $\text{Gal}(\overline{K}/L)$, and put

$$V_K = \text{Ind}_{\text{Gal}(\overline{K}/L)}^{\text{Gal}(\overline{K}/K)}(V_L).$$

For χ a quasi-character of $\mathbb{A}_K^*/K^*$, one has

$$L(V_K, \chi) = L(V_L, \chi \circ N_{L/K}),$$

and similarly

$$\varepsilon(V_K, \chi) = \varepsilon(V_L, \chi \circ N_{L/K}).$$

(D) One also has

$$\varepsilon(V' \oplus V'', \chi) = \varepsilon(V', \chi) \varepsilon(V'', \chi),$$

which makes it possible to define $\varepsilon(V, \chi)$ when V is only a virtual representation.

The assertions (A) and (B) are proved by reduction to the case $\dim V = 1$, via Brauer's theorem. Assertion (C) follows from (B) and from the compatibility of norm and transfer recalled above.

4. Existence of local constants

Theorem 4.1. There exists a unique function ε , satisfying conditions (1) to (4) below, which associates a number

$$\varepsilon(V, \psi, dx) \in \mathbb{C}^*$$

to each isomorphism class of sextuples $(K, \bar{K}, \psi, dx, V, \rho)$, consisting of a local field K , an algebraic closure $\bar{K}$ of K , a nontrivial additive character $\psi : K \rightarrow \mathbb{C}^*$, a Haar measure dx on K , a finite-dimensional complex vector space V , and a representation

$$\rho : W(\bar{K}/K) \longrightarrow \mathrm{GL}(V).$$

- (1) For every exact sequence of representations

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0,$$

one has

$$\varepsilon(V, \psi, dx) = \varepsilon(V', \psi, dx) \varepsilon(V'', \psi, dx).$$

This condition shows that $\varepsilon(V, \psi, dx)$ depends only on the class of V in the Grothendieck group of representations of $W(\bar{K}/K)$, and allows $\varepsilon(V, \psi, dx)$ to be interpreted when V is only a virtual representation.

- (2)

$$\varepsilon(V, \psi, a dx) = a^{\dim V} \varepsilon(V, \psi, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon(V, \psi, dx)$ is independent of dx ; it is denoted $\varepsilon(V, \psi)$.

- (3) If L is a finite separable extension of K in $\bar{K}$, and V_K is the virtual representation of $W(\bar{K}/K)$ induced by a virtual representation of dimension zero V_L of $W(\bar{K}/L)$, then

$$\varepsilon(V_K, \psi) = \varepsilon(V_L, \psi \circ \mathrm{Tr}_{L/K}).$$

- (4) If $\dim V = 1$, and ρ is defined by a quasi-character χ of K^* , then

$$\varepsilon(V, \psi, dx) = \varepsilon(\chi, \psi, dx).$$

4.2. Let

$$v = \sum_{i,j} n_{ij} (H_i, \chi_{ij}) \in R_0^+(W(\bar{K}/K))$$

(1.10). Assume that the H_i are pairwise nonconjugate. Let K_i be the extension of K defined by H_i , and again denote by χ_{ij} the quasi-character of K_i^* defined by χ_{ij} . For any choice of Haar measures dx_i on the K_i , and for ψ an additive character of K , put

$$\varepsilon(v, \psi) = \prod_{i,j} \varepsilon(\chi_{ij}, \psi \circ \mathrm{Tr}_{K_i/K}, dx_i)^{n_{ij}}. \quad (4.2.1)$$

It follows from (3.3.2) that $\varepsilon(v, \psi)$ is independent of the choice of the dx_i .

Lemma 4.3. With the notation 1.9.8, for $v \in R_0^+(W(\bar{K}/K))$,

$$\varepsilon(v, \psi(ax)) = \det(v)(a) \varepsilon(v, \psi).$$

It is enough to check this for $v = (H, \chi) - (H, 1)$, where H corresponds to an extension L of K . Let v_0 be the virtual representation $[\chi] - [1]$ of $W(\bar{K}/L)$. Formula (3.3.3) gives the assertion over L . It is known ([10], XII, §4) that the inclusion of K^* into L^* is identified with the transfer

$$W(\bar{K}/K)^{\mathrm{ab}} \longrightarrow W(\bar{K}/L)^{\mathrm{ab}}.$$

Since $b(v) = \mathrm{Ind}(v_0)$, the lemma follows from 1.2.

Lemma 4.4. The theorem is true in the archimedean case.

For $K \simeq \mathbb{C}$, define ε by (1) and (4), and (2) is automatic. For $K \simeq \mathbb{R}$, one checks from 2.9 that

$$b : R^+(W(\overline{K}/K)) \longrightarrow R(W(\overline{K}/K))$$

is surjective, with kernel generated by the $\chi_s - \chi_t$ where

$$\chi_s = (K^*, \omega_s) - [\omega_s] - [\chi^{-1}\omega_{s+1}].$$

For a virtual representation V , and $v \in R_0^+(W(\overline{K}/K))$ such that

$$b(v) = V - \dim(V)[1],$$

put

$$\varepsilon(V, \psi, dx) = \varepsilon(v, \psi) \varepsilon(1, \psi, dx)^{\dim V}. \quad (4.4.1)$$

We check that the right-hand side is independent of the choice of v , i.e. that, for v satisfying $b(v) = 0$, one has $\varepsilon(v, \psi) = 1$. By 4.3, it is enough to check this for $\psi(x) = \exp(2\pi i x)$. For this choice of ψ , the assertion follows from the preceding determination of $\text{Ker}(b)$, and from the fact that the right-hand sides of (3.4.1), (3.4.2) are independent of s . It is clear that ε defined by (4.4.1) satisfies (1) to (4).

We now restrict to the nonarchimedean case.

4.5. Let K be a nonarchimedean local field, $\overline{K}$ a separable closure of K , and $K_{nr} \subset \overline{K}$ the maximal unramified extension of K . Let J be an open normal subgroup of the inertia group

$$I = \text{Gal}(\overline{K}/K_{nr})$$

and let L be the corresponding extension of K_{nr} . For $\sigma \in I/J$, put

$$t_{I/J}(\sigma) = \inf(v_L(\sigma(x) - x) \mid x \text{ integral in } L),$$

and define functions $a_{I/J}$ and $sw_{I/J}$ by

$$a_{I/J}(\sigma) = -t_{I/J}(\sigma) \quad (\sigma \neq e), \quad (4.5.1)$$

$$sw_{I/J}(\sigma) = 1 - t_{I/J}(\sigma) \quad (\sigma \neq e),$$

$$\sum_{\sigma} a_{I/J}(\sigma) = \sum_{\sigma} sw_{I/J}(\sigma) = 0.$$

By Artin (see [10], VI, §2, th. 1), $a_{I/J}$ is the character of a representation $A_{I/J}$ of I/J , the Artin representation. By loc. cit., prop. 2, $sw_{I/J}$ is likewise the character of a representation $Sw_{I/J}$, the Swan representation. These representations are defined only up to isomorphism. $A_{I/J}$ is the sum of $Sw_{I/J}$ and the augmentation representation of I/J . For $I \supset J \supset K$, one has (loc. cit., prop. 3)

$$(A_{I/K})^{J/K} \xrightarrow{\sim} A_{I/J}, \quad (Sw_{I/K})^{J/K} \xrightarrow{\sim} Sw_{I/J}. \quad (4.5.2)$$

If d is the valuation of the discriminant $\delta_{L/K_{nr}}$, i.e. the valuation of the different $\mathfrak{D}_{L/K}$ ([10], III, §3, prop. 6), and r denotes a regular representation, then (loc. cit., prop. 4)

$$A_{I/K}|_{J/K} = dr_{J/K} + A_{J/K}. \quad (4.5.3)$$

Let V be a representation of $W(\overline{K}/K)$, trivial on $J \subset I$, with character ψ . Define the Artin and Swan conductors of V by

$$a(V) = \dim \text{Hom}_{I/J}(A_{I/J}, V) = sw(V) + \dim V - \dim V^I, \quad (4.5.4)$$

$$sw(V) = \dim \text{Hom}_{I/J}(Sw_{I/J}, V) = \frac{1}{|I/J|} \sum_{\sigma} sw_{I/J}(\sigma) \psi(\sigma).$$

By 4.5.2, these conductors are independent of the choice of J . For V of dimension 1, defined by a quasi-character χ , one has (loc. cit., prop. 5)

$$a(V) = \text{the conductor } a(\chi) \text{ of } \chi. \quad (4.5.5)$$

The preceding notation is therefore compatible with that introduced in 3.4.

With the notation 1.9.8, one has:

Lemma 4.5. For $v \in R_0^+(W(\overline{K}/K))$ and χ an unramified quasi-character of K^* ,

$$\varepsilon(v \cdot \chi, \psi) = \chi(\pi)^{a(v)} \varepsilon(v, \psi).$$

It suffices to verify this for $v = (H, \lambda) - (H, \mu)$, where H defines an extension L of K , and λ, μ are two quasi-characters of L^* . Let d be the valuation of the different of L/K , let f be the residue-degree, let $I \subset W(\overline{K}/K)$ and $J \subset W(\overline{K}/L)$ be the inertia groups, and choose $K \subset J$ small enough. We compute $a(v)$. The adjunction formula for induced representations, (4.5.3), and (4.5.5) give, for every quasi-character ν of H , i.e. of L^* ,

$$a((H, \nu)) = \dim \operatorname{Hom}_{I/K}(A_{I/K}, \operatorname{Ind}_H^W(\nu)) = f \dim \operatorname{Hom}_{J/K}(A_{J/K} + d r_{J/K}, [\nu]|_J) = f a(\nu) + f d. \quad (4.5.1')$$

Thus

$$a(v) = f(a(\lambda) - a(\mu)).$$

For ν a quasi-character of L^* , χ_1 an unramified quasi-character of L^* , and $n = n(\psi \circ \operatorname{Tr}_{L/K})$ as in 3.4, one has by (3.4.3.1) or (3.4.3.2)

$$\frac{\varepsilon(\nu \chi_1, \psi \circ \operatorname{Tr}, dx)}{\varepsilon(\nu, \psi \circ \operatorname{Tr}, dx)} = \chi_1(\pi_L)^{a(\nu)+n}.$$

Hence

$$\varepsilon(v \cdot \chi, \psi) \varepsilon(v, \psi)^{-1} = \chi \circ N_{L/K}(\pi_L^{a(\lambda)-a(\mu)}) = \chi(\pi)^{f(a(\lambda)-a(\mu))},$$

which is the announced formula.

Terminology 4.6. Let K_1 be a Galois extension, finite or not, of K , with Galois group G . Let α be a quasi-character of K^* , and ψ a nontrivial additive character of K . We shall say that there exists an α, ψ -theory of constants for K_1/K if there exists $\varepsilon_{\alpha, \psi}$ of the following type:

- (0) For H an open subgroup of G , defining a finite extension L of K , for V a representation of H , and dx a Haar measure on L ,

$$\varepsilon_{\alpha, \psi}(V, dx) \in \mathbb{C}^*$$

is defined.

- (1) For every exact sequence of representations

$$0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0,$$

one has

$$\varepsilon_{\alpha, \psi}(V, dx) = \varepsilon_{\alpha, \psi}(V', dx) \varepsilon_{\alpha, \psi}(V'', dx).$$

Thus $\varepsilon_{\alpha, \psi}$ still makes sense for virtual representations.

- (2) One has

$$\varepsilon_{\alpha, \psi}(V, a dx) = a^{\dim V} \varepsilon_{\alpha, \psi}(V, dx).$$

In particular, for virtual V of dimension 0, $\varepsilon_{\alpha, \psi}(V, dx)$ is independent of dx ; it is denoted $\varepsilon_{\alpha, \psi}(V)$.

- (3) For $H_1 \subset H_2 \subset G$, defining $L_1 \supset L_2$, and V a virtual representation of dimension 0 of H_1 , one has

$$\varepsilon_{\alpha, \psi}(\text{Ind}_{H_1}^{H_2} V) = \varepsilon_{\alpha, \psi}(V).$$

- (4) For V of dimension 1, defined by a character χ of L^* ,

$$\varepsilon_{\alpha, \psi}(V, dx) = \varepsilon(\chi \alpha \circ N_{L/K}, \psi \circ \text{Tr}_{L/K}, dx).$$

Lemma 4.7. Let K_1/K and α be as above, and let ψ be a nontrivial additive character of K . There exists at most one α, ψ -theory of constants for K_1/K . For one to exist, it is necessary and sufficient that, for every

$$v \in \text{Ker}(R_0^+(G) \rightarrow R(G)),$$

one have

$$\varepsilon(v \cdot \alpha, \psi) = 1.$$

Let $H \subset G$, L , and V be as above. By 1.5, there exists $v \in R_0^+(H)$ whose image in $R(H)$ is $V - \dim(V)[1]$. Necessarily

$$\varepsilon_{\alpha, \psi}(V, dx) = \varepsilon(v \cdot (\alpha \circ N_{L/K}), \psi \circ \text{Tr}_{L/K}) \varepsilon(\alpha \circ N_{L/K}, \psi \circ \text{Tr}_{L/K}, dx)^{\dim V}, \quad (4.7.1)$$

which gives uniqueness. The hypothesis ensures that $\varepsilon_{\alpha, \psi}$, defined by this formula, is independent of the choice of v ; one easily checks axioms (0) to (4).

Lemma 4.8.

- (i) Let β be an unramified quasi-character of K^* , respectively let $\psi' = \psi(ax)$ be a nontrivial additive character. For there to exist an α, ψ -theory of constants for K_1/K , it is necessary and sufficient that there exist an $\alpha\beta, \psi'$ -theory, respectively an α, ψ' -theory. For H, L, V as above,

$$\psi_L = \psi \circ \text{Tr}_{L/K}, \quad \beta_L = \beta \circ N_{L/K}, \quad \alpha_L = \alpha \circ N_{L/K},$$

one then has

$$\varepsilon_{\alpha\beta, \psi}(V, dx) = \beta_L \left(\pi^{a(V\alpha_L) + \dim(V)n(\psi_L)} \right) \varepsilon_{\alpha, \psi}(V, dx), \quad (4.8.1)$$

$$\varepsilon_{\alpha, \psi'}(V, dx) = \|a\|_L^{-\dim(V)} \det(V\alpha_L)(a) \varepsilon_{\alpha, \psi}(V, dx). \quad (4.8.2)$$

The first assertion follows from criterion 4.7 and from 4.5, respectively from 4.3. It is enough to prove (4.8.1) and (4.8.2) for $K = L$. Moreover, one may suppose either that $V \in R^0(\text{Gal}(K_1/K))$, or that $V = [1]$. In the first case one applies 4.5 and 4.3. In the second, one uses (4.7.1) and the definition 3.4.3.

We shall say that there exists an α -theory of constants for K_1/K if, for every nontrivial additive character ψ , or for one such ψ , there exists an α, ψ -theory of constants for K_1/K . For $K_1 \supset L \supset K$, there is then an $\alpha \circ N_{L/K}$ -theory of constants for K_1/L .

Reduction 4.9. Theorem 4.1 is implied by the following assertion:

- (*) for every nonarchimedean local field K and every finite Galois extension K_1/K ,

there exists an unramified quasi-character α of K^* and an α -theory of constants for K_1/K .

By 4.8, assertion (*) implies that, for $\overline{K}$ a separable closure of K , and for any unramified quasi-character α of K^* , there exists an α -theory of constants for $\overline{K}/K$. The discussion below will allow us to pass from the Galois group to the Weil group.

4.10. Every representation V of $W(\overline{K}/K)$ is trivial on a suitable finite-index subgroup J of inertia I . Let F be a geometric Frobenius. Since I/J is finite, a suitable power F^n of F acts trivially on I/J by conjugation, and hence is central in $W(\overline{K}/K)/J$. We shall say that V has a *type* if a power F^m of F^n has only one eigenvalue a . The type of V is then the element of the inductive limit

$$\varinjlim_{n|m} \{X_n, \varphi_{n,m}\}, \quad X_n = \mathbb{C}^*, \quad \varphi_{n,m}(x) = x^{m/n},$$

defined by $a \in X_m = \mathbb{C}^*$. Every irreducible representation has a type.

The representations of $W(\overline{K}/K)$ of a given type τ form an abelian category R_τ , and the category of all representations of $W(\overline{K}/K)$ is the sum of these subcategories. If $R_\tau(W(\overline{K}/K))$ is the Grothendieck group of R_τ , one therefore has

$$R(W(\overline{K}/K)) = \bigoplus_{\tau} R_\tau(W(\overline{K}/K)). \quad (4.10.1)$$

The semisimple representations of type 1 are exactly the continuous representations of $\text{Gal}(\overline{K}/K)$. If X_τ is a quasi-character of type τ , and there are quasi-characters, even unramified ones, of all types, then twisting by X_τ defines an isomorphism

$$\otimes X_\tau : R(\text{Gal}(\overline{K}/K)) \xrightarrow{\sim} R_\tau(W(\overline{K}/K)). \quad (4.10.2)$$

Let L be a finite separable extension of K in $\overline{K}$, and let f be the residue-degree. If a representation V of $W(\overline{K}/L)$ has type σ , the representation induced to $W(\overline{K}/K)$ has type σ^f . We say that V is of K -type τ if $\sigma^f = \tau$. The set of representations of $W(\overline{K}/L)$ of a given K -type is therefore stable under induction.

Variante 4.10.3. The preceding arguments apply to every group which is an extension of $\mathbb{Z}$ by a finite group, and to projective limits of such extensions. The case of global Weil groups will be useful later.

4.11. Assume (*). For every type τ , let X_τ be an unramified character of type τ . Since there exists an X_τ -theory of constants, one can define ε on $R_\tau(W(\overline{K}/K))$ by

$$\varepsilon(V, \psi, dx) = \varepsilon_{X_\tau, \psi}(V \otimes X_\tau^{-1}, dx).$$

It follows from the uniqueness 4.7 that this ε is independent of the choice of X_τ . One then defines ε on $R(W(\overline{K}/K))$ by additivity (4.10.1). By 4.7, if a theory 4.1 exists, it is given by this ε .

We have arranged things so that ε satisfies (1), (2), (4), and (3) when V_L has a type. It remains to verify (3) for V_L of the form

$$[\lambda \circ N_{L/K}] - [\mu \circ N_{L/K}],$$

where λ and μ are two unramified quasi-characters of K^* , of given types. By (3.4.3.1), (3.3.2), and (3.3.3), one has

$$\varepsilon([\lambda \circ N_{L/K}] - [\mu \circ N_{L/K}], \psi \circ \text{Tr}_{L/K}) = (\lambda\mu^{-1}) \circ N_{L/K} \left(\pi_L^{n(\psi \circ \text{Tr}_{L/K})} \right).$$

On the other hand, by 4.8,

$$\varepsilon(\text{Ind}([\lambda \circ N_{L/K}] - [\mu \circ N_{L/K}]), \psi) = (\lambda\mu^{-1}) \left(\pi_K^{a(\text{Ind}(\mu \circ N_{L/K})) + [L:K] n(\psi)} \right).$$

Let d be the valuation of the different of L/K , e the ramification index, and f the residue-degree. Formula (4.5.1) gives

$$a(\text{Ind}(\mu \circ N_{L/K})) = fd.$$

The formula to be proved is therefore reduced to

$$f n(\psi \circ \text{Tr}_{L/K}) = fd + [L:K] n(\psi),$$

that is,

$$n(\psi \circ \text{Tr}_{L/K}) = d + e n(\psi),$$

which is essentially the definition of the different ([10], III, §3). The key point in the proof of (*) is the following lemma.

Lemma 4.12. Let K'/K be a Galois extension of global fields. For every place w of K , choose a place of K' above it and denote it again by w . Let α be a quasi-character of the idèle-class group of K , and let v be a place of K which does not decompose in K' . If, for every finite place $w \neq v$ of K , there exists an α_w -theory of constants for K'_w/K_w , then there exists an α_v -theory of constants for K'_v/K_v .

Let ψ be a nontrivial character of $\mathbb{A}_K/K$, and let ψ_v be its v -component. We shall construct an α_v, ψ_v -theory of constants for K'_v/K_v .

One has

$$\text{Gal}(K'/K) = \text{Gal}(K'_v/K_v).$$

A subgroup H of $\text{Gal}(K'_v/K_v)$ therefore defines, besides an extension L_v of K_v , an extension L of K with completion L_v . Let dx be the Tamagawa measure on $\mathbb{A}_L$, so that $\mathbb{A}_L/L$ has volume 1, and write

$$dx = \otimes_w dx_w,$$

with dx_v chosen in advance. Every representation V_v of H defines a representation V of $\text{Gal}(K'/L)$, and, by restriction, representations V_w of the decomposition groups $\text{Gal}(K'_w/K_w)$. A global constant $\varepsilon(V, \alpha)$ was defined in 3.12.

Let w be a place of L , and again write w for its image in K . For w infinite, one has by 4.4 local constants

$$\varepsilon(V_w, (\alpha \circ N_{L/K})_w, (\psi \circ \text{Tr}_{L/K})_w, dx_w).$$

For w finite and different from v , one also has, by hypothesis, constants

$$\varepsilon_{\alpha_w, \psi_w}(V_w, dx_w).$$

Denoting both kinds by $\varepsilon_\alpha(V_w, \psi_w, dx_w)$, put

$$\varepsilon_{\alpha_v, \psi_v}(V_v, dx_v) = \frac{\varepsilon(V, \alpha)}{\prod_{w \neq v} \varepsilon_\alpha(V_w, \psi_w, dx_w)}.$$

One checks that $\varepsilon_{\alpha_v, \psi_v}$ does not depend on the choice of the decomposition $dx = \otimes dx_w$, and satisfies axioms (0) to (4). The key point is 3.12(C).

Lemma 4.13. Let K'/K be a Galois extension of local fields. There exists a Galois extension of global fields ${}^1K'/{}^1K$, and a place v of 1K which does not decompose in ${}^1K'$, such that K is identified with the completion of 1K at v , and K' with a subextension of ${}^1K'_v/{}^1K_v$.

This is trivial: first take ${}^1K'/{}^1K$ with ${}^1K'_v \simeq K' \supset K = {}^1K_v$, and then replace 1K by the extension defined by a decomposition group at v .

Lemma 4.14. Let K be a global field, let S be a finite set of finite places of K , and let $m : S \rightarrow \mathbb{N}$. There exists a character α of the idèle-class group such that, for $w \in S$, the conductor of α_w is $\geq m(w)$, and such that, for every finite place outside S , α_w is unramified.

If K is a number field, one can indeed freely specify α on

$$\prod_{w \text{ finite}} \mathcal{O}_w^*,$$

because this compact subgroup of $\mathbb{A}^*$ meets K^* only in $\{1\}$. If K is a function field, with field of constants k , one has

$$\prod_w \mathcal{O}_w^* \cap K^* = k^*,$$

and one observes that there exist characters of $\mathcal{O}_w^*$, of arbitrarily large conductor, which are trivial on k^* .

By 4.12, 4.13, and 4.14, assertion 4.9 (*), and hence 4.1, follow from the following two lemmas, where K'/K is a finite Galois extension of nonarchimedean local fields, with Galois group G , and α is a quasi-character of K^* .

Lemma 4.15. If α and K'/K are unramified, there exists an α -theory of constants for K'/K .

Choose for ψ a character of K such that $n(\psi) = 0$. It is then enough to take

$$\varepsilon_{\alpha, \psi}(V, dx) = \left(\int_L dx \right)^{\dim(V)}.$$

Lemma 4.16. For K'/K fixed, there exists m such that, if the conductor of α is $\geq m$, there exists an α -theory of constants for K'/K .

Let α have conductor m , and put

$$n = \left\lfloor \frac{m+1}{2} \right\rfloor.$$

Then the function $\alpha(1+a)$ is additive in a for $v(a) \geq m-n$. If ψ is a nontrivial additive character of K , there is therefore a y , of valuation $-(m+n(\psi))$, well-defined modulo $\pi^{-(n+n(\psi))}$, such that

$$\alpha(1+a) = \psi(ya) \quad \text{for } v(a) \geq n.$$

For χ a quasi-character of K^* , of conductor $\leq m - n$, one then has

$$\varepsilon(\chi\alpha, \psi, dx) = \sum_{u \in \mathcal{O}^*/(1+(\pi^n))} \int_{\pi^{-m-n(\psi)}u} \chi^{-1}(x)\alpha^{-1}(x)\psi(x) dx. \quad (4.16.1)$$

For $\pi^{-m-n(\psi)}u$ not contained in $y(1 + (\pi^{m-n}))$, the corresponding partial integral is zero: it is the integral of a nontrivial additive character. Since χ is constant with value $\chi(y)$ on $y(1 + (\pi^{m-n}))$, one has

$$\varepsilon(\chi\alpha, \psi, dx) = \chi^{-1}(y) \varepsilon(\alpha, \psi, dx).$$

We shall show that, for m sufficiently large, one obtains an α, ψ -theory of constants for K'/K by putting, for $H \subset G$ defining an extension L of K , and V a representation of H ,

$$\varepsilon_{\alpha, \psi}(V, dx) = \det(V)^{-1}(y) \varepsilon(\alpha \circ N_{L/K}, \psi \circ \text{Tr}_{L/K}, dx)^{\dim V}.$$

Axioms (0), (1), and (2) are trivial. Axiom (3) follows from the fact that the inclusion of K^* in L^* is identified with the transfer

$$W(\overline{K}/K)^{\text{ab}} \longrightarrow W(\overline{K}/L)^{\text{ab}}$$

([10], XIII, §4), and from 1.2. It remains to prove axiom (4).

Let $H \subset G$ correspond to L/K , and let e be the ramification index. Recall that, for $u \in \mathcal{O}_L$ and for n the least integer such that $ne \geq 2v(u)$,

$$N_{L/K}(1+u) = 1 + \text{Tr}_{L/K}(u) \pmod{\pi^n}. \quad (4.16.2)$$

This is checked by writing the norm of $1+u$ as the product of its conjugates. We shall write $O(1)$ for any number bounded in terms of K'/K alone. By 4.16.2, if α has conductor m , then $\alpha \circ N_{L/K}$ has conductor $em - O(1)$. Moreover, for

$$\frac{v(a)}{e} \geq \left\lfloor \frac{m+1}{2} \right\rfloor + O(1),$$

one has

$$\alpha \circ N_{L/K}(1+a) = \alpha(1 + \text{Tr}(a)) = \psi(y \text{Tr } a) = \psi \circ \text{Tr}_{L/K}(ya).$$

For every character χ of H , of conductor $O(1)$, one can repeat the reasoning used above, obtaining the desired formula

$$\varepsilon(\chi\alpha \circ N_{L/K}, \psi \circ \text{Tr}_{L/K}, dx) = \chi^{-1}(y) \varepsilon(\alpha \circ N_{L/K}, \psi \circ \text{Tr}_{L/K}, dx).$$

5. Formula list

5. Formula list.

(5.1) Besides the local constant $\varepsilon(V, \psi, dx)$ defined in 4.1, we shall also consider, in the nonarchimedean case, the constant ε_0 defined by

$$\varepsilon(V, \psi, dx) = \varepsilon_0(V, \psi, dx) \det(-F, V^I)^{-1}.$$

The constants ε and ε_0 have the following properties.

(5.2) ε and ε_0 are additive in V , and hence still make sense for virtual V .

(5.3) ε and ε_0 are independent of dx for virtual V of dimension 0. They are denoted $\varepsilon(V, \psi)$ and $\varepsilon_0(V, \psi)$. More precisely,

$$\varepsilon(V, \psi, a dx) = a^{\dim V} \varepsilon(V, \psi, dx),$$

and similarly for ε_0 .

(5.4) One has

$$\varepsilon(V, \psi(ax), dx) = \det(V)(a) \|a\|^{-\dim V} \varepsilon(V, \psi, dx),$$

and similarly for ε_0 .

(5.5) For K nonarchimedean and χ an unramified quasi-character of K^* , one has

$$\varepsilon(V\chi, \psi, dx) = \chi\left(\pi^{a(V)+\dim(V)n(\psi)}\right) \varepsilon(V, \psi, dx), \quad (5.5.1)$$

and

$$\varepsilon_0(V\chi, \psi, dx) = \chi\left(\pi^{\dim(V)+sw(V)+\dim(V)n(\psi)}\right) \varepsilon_0(V, \psi, dx).$$

In particular, if

$$\varepsilon(V, \psi, dx, s) = \varepsilon(V\omega_s, \psi, dx),$$

and similarly for ε_0 , then

$$\varepsilon(V, \psi, dx, s) = \varepsilon(V, \psi, dx) q^{-(a(V)+\dim(V)n(\psi))s}, \quad (5.5.2)$$

and

$$\varepsilon_0(V, \psi, dx, s) = \varepsilon_0(V, \psi, dx) q^{-(sw(V)+\dim(V)(n(\psi)+1))s}.$$

For W an unramified representation, one similarly has

$$\varepsilon(V \otimes W, \psi, dx) = \det W \left(\pi^{a(V)+\dim(V)n(\psi)} \right) \varepsilon(V, \psi, dx)^{\dim W}, \quad (5.5.3)$$

$$\varepsilon_0(V \otimes W, \psi, dx) = \det W \left(\pi^{sw(V)+\dim(V)(n(\psi)+1)} \right) \varepsilon_0(V, \psi, dx)^{\dim W}.$$

(5.6) For L/K a finite separable extension, V_L a virtual representation of dimension 0 of $W(\overline{K}/L)$, and V_K the representation induced to $W(\overline{K}/K)$, one has

$$\varepsilon(V_K, \psi) = \varepsilon(V_L, \psi \circ \text{Tr}_{L/K}), \quad (5.6.1)$$

and similarly for ε_0 . In other words, there exists

$$\lambda(L/K, \psi, dx_L, dx_K) \in \mathbb{C}^*$$

such that, for V_L of arbitrary dimension,

$$\varepsilon(V_K, \psi, dx_K) = \lambda(L/K, \psi, dx_L, dx_K)^{\dim(V_L)} \varepsilon(V_L, \psi \circ \text{Tr}_{L/K}, dx_L). \quad (5.6.2)$$

The same formula holds for ε_0 .

In Langlands [9], it is (5.6.2) which provides the framework of the theory. Moreover, he systematically takes dx to be the Haar measure self-dual for ψ , and omits it from the notation. The constants are also normalized to have complex absolute value 1. All of this forces on him another formula (5.4).

5.7. Let V^* be the dual of V , and let dx' be the Haar measure dual to dx , relative to ψ . One has

$$\varepsilon(V, \psi, dx) \varepsilon(V^* \omega_1, \psi(-x), dx') = 1. \quad (5.7.1)$$

In particular, if K is nonarchimedean and V is unitary, so that $V^* = \overline{V}$, one has

$$|\varepsilon(V, \psi, dx)|^2 = \left(\frac{dx'}{dx} \right)^{-\dim(V)} q^{a(V)+\dim(V)n(\psi)}. \quad (5.7.2)$$

5.8. ε^{-1} appears in Tate's local functional equation; putting $\chi' = \omega_1 \chi^{-1}$,

$$\frac{\int \widehat{f}(x) \chi'(x) d^*x}{L(\chi')} = \varepsilon(\chi, \psi, dx) \frac{\int f(x) \chi(x) d^*x}{L(\chi)} \quad (\text{Tate}). \quad (5.8.1)$$

For K nonarchimedean, and $\gamma \in K^*$ of valuation $1 + sw(\chi) + n(\psi)$, (3.4.3.4) gives

$$\varepsilon_0(\chi, \psi, dx) = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x) \psi(x) dx. \quad (5.8.2)$$

5.9. For K nonarchimedean, if χ is unramified, one has

$$\varepsilon(\chi, \psi, dx) = \chi(\pi^{n(\psi)}) q^{n(\psi)} \int_{\mathcal{O}} dx.$$

In particular, if $n(\psi) = 0$ and $\int_{\mathcal{O}} dx = 1$, then

$$\varepsilon(\chi, \psi, dx) = 1.$$

5.10. Take K nonarchimedean, $a(\chi) \leq 1$, $n(\psi) = -1$, and

$$\int_{(\pi)} dx = 1.$$

Let χ_k and ψ_k be the characters of k^* and k defined by χ and ψ . One has

$$\varepsilon_0(\chi, \psi, dx) = -\tau(\chi_k, \psi_k), \quad (5.10.1)$$

where τ is the Gauss sum

$$\tau(\chi_k, \psi_k) = - \sum_{x \in k^*} \chi_k^{-1}(x) \psi_k(x), \quad (5.10.2)$$

equal to 1 for trivial χ_k . In particular,

$$\varepsilon_0([\chi] - [1], \psi) = \tau(\chi_k, \psi_k). \quad (5.10.3)$$

5.11. Let K be a global field. For V a representation of the global Weil group of K , ψ a nontrivial additive character of $\mathbb{A}_K/K$, and

$$dx = \bigotimes_v dx_v$$

the Tamagawa measure on $\mathbb{A}_K$, put

$$L(V) = \prod_v L(V_v), \quad (5.11.1)$$

$$\varepsilon(V) = \prod_v \varepsilon(V_v, \psi_v, dx_v). \quad (5.11.2)$$

The global functional equation is written

$$L(V) = \varepsilon(V) L(V^* \otimes \omega_1). \quad (5.11.3)$$

Proofs. Formulas 5.2, 5.3, 5.6, and 5.8 are 4.1, respectively (1), (2), (3), (4), together with 3.3. Formulas 5.4 and 5.5 follow from 4.8. It is enough to prove (5.7.1) for V defined by a character; the formula then follows from (5.8.1) and from the Fourier inversion formula. Formula (5.7.2) follows from (5.7.1), (5.3), and (5.4). Formula 5.9 follows from (3.4.3.1), and 5.10 from (5.8.2). Finally, 5.11 follows from the preceding facts and from (3.11.4), by known arguments [1], [19].

The restriction of 4.1 to the case of extensions and tamely ramified characters is equivalent to the following identities on Gauss sums (5.10.2). Here k denotes a finite field with q elements, and ψ a nontrivial additive character of k .

5.12. Hasse–Davenport. Let k' be an extension of k , of degree n , and let χ be a character of k^* . One has

$$\tau(\chi \circ N_{k'/k}, \psi \circ \text{Tr}_{k'/k}) = \tau(\chi, \psi)^n. \quad (5.12.1)$$

5.13. Let χ be a character of k^* , and let n be an integer prime to q . Let X be a set of representatives of the isomorphism classes of pairs (k', χ') , consisting of an extension k' of k and a character χ' of k'^* , such that

$$\chi'(x^n) = \chi \circ N_{k'/k}(x), \quad (\text{a})$$

and

$$\chi' \text{ is not of the form } \chi'' \circ N_{k'/k''} \quad (\text{b})$$

for k'' intermediate between k and k' , i.e.

$$\chi'^{q^{f'}} \neq \chi' \quad (0 < f' < [k' : k]).$$

Put

$$\tau(\chi, \psi^n) = \lambda_n(\chi, \psi) \prod_{(k', \chi') \in X} \tau(\chi', \psi \circ \text{Tr}_{k'/k}).$$

Then

$$\lambda_n(\chi, \psi) \text{ is independent of } \chi. \quad (5.13.1)$$

This identity splits into the following two identities.

5.14 (Langlands [9], 7.8). Let ℓ be a prime number prime to q , let f be the order of q in $(\mathbb{Z}/\ell)^*$, let k' be an extension of k of degree f , let T be a set of representatives of the $\text{Gal}(k'/k)$ -orbits in the nontrivial characters of order ℓ of k'^* , and let χ be a character of k^* . One has

$$\tau(\chi^\ell, \psi^\ell) \prod_{\mu \in T} \tau(\mu, \psi \circ \text{Tr}_{k'/k}) = \tau(\chi, \psi) \prod_{\mu \in T} \tau(\chi \circ N_{k'/k} \cdot \mu, \psi \circ \text{Tr}_{k'/k}).$$

5.15 (Langlands [9], 7.9). Let ℓ be a prime number dividing $q - 1$, let T be the set of characters of order ℓ of k^* , let χ be a character of k^* which is not an ℓ -th power, let k' be an extension of degree ℓ of k , and let χ' be a character of k'^* such that

$$\chi'(x^\ell) = \chi \circ N_{k'/k}(x).$$

One has

$$\tau(\chi, \psi^\ell) \prod_{\mu \in T} \tau(\mu, \psi) = \tau(\chi', \psi \circ \text{Tr}_{k'/k}).$$

5.16. Proofs. If K'/K is a tamely ramified extension of nonarchimedean local fields, and if ψ is an additive character of K such that

$$n(\psi) = -1,$$

then still

$$n(\psi \circ \text{Tr}_{K'/K}) = -1,$$

so that 5.10 applies both to K and to K' .

Let V be a tamely ramified representation of $W(\overline{K}/K)$. One can always write it as a sum of representations induced by tame characters χ_i of unramified extensions K_i/K :

$$V = \dim(V)[1] + \sum_i \text{Ind}([\chi_i] - [1]) + \sum_i (\text{Ind}([1]) - [K_i : K][1]).$$

Let k_i be the residue field of K_i , and let $\tilde{\chi}_i$ be the character of k_i^* defined by χ_i . For dx a Haar measure on K such that

$$\int_{(\pi)} dx = 1,$$

one finds

$$\varepsilon_0(V, \psi, dx) = (-1)^{\dim V} \prod_i \tau(\tilde{\chi}_i, \psi_k \circ \text{Tr}_{k_i/k}). \quad (5.16.1)$$

Formula (5.12) is obtained by taking an unramified extension K'/K , and by writing that, for χ a character, here tamely ramified, of K^* , one has

$$\text{Ind}(\chi \circ N_{K'/K} - [1]) = \sum_{\mu} ([\chi\mu] - [\mu]),$$

where μ runs through the unramified characters of order dividing $[K' : K]$. This formula (5.12) ensures that the right-hand side of (5.16.1) depends only on V .

The identities 5.13 are obtained by taking a totally ramified extension K'/K of degree n , and by combining 5.6.1 and (5.16.1) for V induced by a tame character χ of K' . Formulas (5.14) and (5.15) are the special case of (5.13.1) where n is a prime number ℓ . In 5.14, respectively 5.15, one supposes that the character χ of K'^* is, respectively is not, an ℓ -th power.

6. Reduction modulo ℓ of local constants

Let K be a nonarchimedean local field, let $\overline{K}$ be a separable closure of K , and resume the notation of 2.2. Let A be a commutative ring in which p is invertible. The exposition would be simplified, without loss of generality, by assuming A local, or even a field. For $x \in K$, $\|x\|$ has a meaning in A by hypothesis.

6.1. A Haar measure μ on K , with values in a $\mathbb{Z}[1/p]$ -module M , is a finitely additive function on compact open subsets of K , with values in M , invariant under translation. The additive function defined by

$$\mu_0(a + x\mathcal{O}) = \|x\|$$

is the unique Haar measure with values in $\mathbb{Z}[1/p]$ such that

$$\mu_0(\mathcal{O}) = 1,$$

and every Haar measure with values in M is of the form

$$m\mu_0 \quad (m = \mu(\mathcal{O}) \in M).$$

One can integrate a locally constant compactly supported function, with values in A , against a Haar measure with values in A .

6.2. Let

$$\rho : W(\overline{K}/K) \longrightarrow \text{GL}_A(V)$$

be a representation of the Weil group on a free, or projective, A -module V , and let J be an open normal subgroup of I on which ρ is trivial. For every prime number $\ell \neq p$, the Swan representation $Sw_{I/J}$ can be realized as a projective $\mathbb{Z}_\ell[I/J]$ -module. If A is a $\mathbb{Z}_\ell$ -algebra, then

$$\text{Hom}_{I/J}(Sw_{I/J}, V)$$

is a projective A -module. If A is also local, it is free; its rank is the Swan conductor of V . More generally, one checks that the Swan conductor of V can always be defined as a locally constant integer-valued function on $\text{Spec}(A)$. This conductor is invariant under extension of the scalar ring A .

6.3. If ζ is a p^n -th root of unity in A , the order of ζ is a locally constant function on $\text{Spec}(A)$. Indeed, any two of the cyclotomic polynomials $\Phi_{p^i}(X)$ generate the unit ideal in $A[X]$, since $p \in A^*$. Thus A is a product of rings A_i , $1 \leq i \leq n$, the i -th coordinate of ζ satisfying

$$\Phi_{p^i}(\zeta) = 0;$$

on $\text{Spec}(A_i)$, ζ has order p^i .

If ψ is a continuous additive character of K with values in A^* , then it takes its values in the p^n -th roots of unity. This makes it possible to define $n(\psi)$ as a locally constant function on $\text{Spec}(A)$: at a prime ideal λ , $n(\psi)$ is $+\infty$ if ψ is trivial modulo λ , and otherwise the largest integer n such that

$$\psi|_{\pi^{-n}\mathcal{O}}$$

is trivial modulo λ , or in the localization A_λ , which is the same thing. By abuse of language, one says that ψ is nontrivial if $n(\psi)$ is nowhere $+\infty$.

6.4. For ψ a nontrivial additive character of K with values in A^* , for $\chi \in \text{Hom}(K^*, A^*)$, and for dx a Haar measure with values in A , one can now define

$$\varepsilon_0(\chi, \psi, dx) \in A^*$$

by formula (3.4.3.4). More precisely, if $\text{Spec}(A)$ is not connected, one first writes

$$A = \prod_i A_i$$

with $n(\psi)$ and $Sw(\chi)$ constant on $\text{Spec}(A_i)$. The i -th projection of ε_0 is then defined by (3.4.3.4).

Theorem 6.5. There exists a unique function ε_0 of the following type:

- (a) For K a nonarchimedean local field, $\overline{K}$ a separable closure of K , A a field of characteristic $\neq p$, V a representation of $W(\overline{K}/K)$ in a finite-dimensional vector space over A , ψ a nontrivial additive character with values in A^* , and dx a nonzero Haar measure with values in A , one has

$$\varepsilon_0(V, \psi, dx) \in A^*.$$

- (b) ε_0 is invariant under extension of the field A .

- (c) Let A be a discrete valuation ring of residual characteristic $\neq p$, let F be its fraction field, and let $s = A/\mathfrak{m}$ be its residue field. For V a representation of $W(\overline{K}/K)$ in a finite-rank free A -module, and for ψ and dx with values in A , one has

$$\varepsilon_0(V_F, \psi, dx) \in A^*,$$

and its reduction modulo $\mathfrak{m}$ is

$$\varepsilon_0(V_s, \overline{\psi}, \overline{dx}).$$

- (d) Conditions (1) to (3) of 4.1 are satisfied.

- (e) For $\dim V = 1$, ε_0 is given by 6.4.

By Galois descent, it is enough to treat the case where A is separably closed. If A has characteristic 0, the assertion, being purely algebraic, follows from the case $A = \mathbb{C}$ treated in 4.1. Thus suppose A has characteristic $\ell \neq 0$.

Let G be a finite quotient of $\text{Gal}(\overline{K}/K)$, and let A_0 be a discrete valuation ring of unequal characteristic, with residue field $A_0/\mathfrak{m} = A$. Assume that ψ has been lifted to an additive character with values in A_0^* , that dx has been lifted to a Haar measure with values in A_0 , and that the fraction field E of A_0 is large enough, in the sense of 1.4, relative to G .

For every character χ of a subgroup of G with values in E^* , in fact in A_0^* , the corresponding constant ε_0 is clearly in A_0 . By 5.7.1, it is even in A_0^* . It follows that, for every representation V_E of G over E ,

$$\varepsilon_0(V_E, \psi, dx) \in A_0^*.$$

The decomposition homomorphism

$$d : R_E(G) \longrightarrow R_A(G)$$

is known to be surjective: virtual representations lift. For $V \in R_A(G)$, choose $\tilde{V} \in R_E(G)$ with $d(\tilde{V}) = V$, and put

$$\varepsilon_0(V, \overline{\psi}, \overline{dx}) = \text{the reduction modulo } \mathfrak{m} \text{ of } \varepsilon_0(\tilde{V}, \psi, dx).$$

By 1.8, to justify this definition it is enough to check that:

$$(*) \quad \text{for } H \subset G \text{ defining an extension } L \text{ of } K,$$

$$\psi_L = \psi \circ \text{Tr}_{L/K}, \quad \chi', \chi'' \text{ characters of } H \text{ with values in } A_0^*,$$

if χ' and χ'' are congruent modulo $\mathfrak{m}$, then

$$\begin{aligned} \varepsilon_0\left(\text{Ind}_H^G([\chi'] - [\chi'']), \psi\right) &= \varepsilon_0([\chi'] - [\chi''], \psi_L) \\ &\equiv 1 \pmod{\mathfrak{m}}. \end{aligned}$$

The Swan conductors of χ' and χ'' are both equal to the Swan conductor of χ' modulo $\mathfrak{m}$. For

$$y = \pi^{1+Sw(\chi')+n(\psi_L)},$$

the formulas of 3.4.3 give

$$\varepsilon_0(\chi', \psi_L, dx_L) = \int_{y^{-1}\mathcal{O}_L^*} \chi'^{-1}(x) \psi_L(x) dx_L,$$

and similarly for χ'' . The numerator and denominator are units, and are clearly congruent modulo $\mathfrak{m}$. Hence

$$\frac{\varepsilon_0(\chi', \psi_L, dx_L)}{\varepsilon_0(\chi'', \psi_L, dx_L)} \equiv 1 \pmod{\mathfrak{m}}.$$

For V a representation of $\text{Gal}(\overline{K}/K)$, one therefore obtains a function ε_0 with the announced properties. Its formal properties are the same as those made explicit in the formula list 5.1 in the case $A = \mathbb{C}$. To pass from $\text{Gal}(\overline{K}/K)$ to $W(\overline{K}/K)$, one may therefore repeat the argument used in 4.9, with the help of the second formula (5.5.1).

Remark 6.6. Throughout this section it was essential to work with ε_0 rather than with ε . First, this permits the use of the Swan conductor, which is invariant under reduction modulo ℓ , rather than the Artin conductor; second, it avoids having to distinguish, in the definition of $\varepsilon_0(\chi, \psi, dx)$, between the unramified and tamely ramified cases. The meaning of ε_0 will appear more clearly in the next section.

7. Modular L -functions

7.1. Let X be a nonsingular projective curve over $\mathbb{F}_p$. Assume X irreducible, let K be its field of rational functions, and identify the closed points of X with the places of K . We do not assume that X is geometrically irreducible, i.e. that $\mathbb{F}_p$ is algebraically closed in K .

7.2. Let A be a noetherian ring. It is known what a constructible sheaf of A -modules on $X_{\text{ét}}$ is (SGA 4, IX, 2.3). For every $x \in X$, and for $k(\overline{x})$ a separably closed extension of $k(x)$, such a sheaf $\mathcal{G}$ has a fiber

$$\mathcal{G}_{\overline{x}}$$

at x , which is a finite-type A -module. It depends only on the separable closure of $k(x)$ in $k(\overline{x})$, and $\text{Gal}(\overline{k(x)}/k(x))$ acts on $\mathcal{G}_{\overline{x}}$ by transport of structure.

Moreover, for v a place of K , i.e. a closed point of X , and $\overline{K}_v$ an algebraic closure of K_v , there is a specialization map

$$\text{sp}_v : \mathcal{G}_{\overline{k(v)}} \longrightarrow \mathcal{G}_{\overline{K}_v}.$$

This map is equivariant; its image is therefore contained in $\mathcal{G}_{\overline{K}_v}^{I_v}$. We say that $\mathcal{G}$ is unramified at v if sp_v is an isomorphism. A constructible sheaf is unramified almost everywhere.

Once a separable closure $\overline{K}$ of K , and a place of $\overline{K}$ above each place of K , have been chosen, a constructible sheaf of A -modules can conversely be defined as follows:

- (a) one gives the fibers $\mathcal{G}_{\overline{K}}$ and $\mathcal{G}_{\overline{k(v)}}$, respectively $A\text{-Gal}(\overline{K}/K)$ -modules or $A\text{-Gal}(\overline{k(v)}/k(v))$ -modules, of finite type over A ;
- (b) one gives the specialization maps, equivariant for the action of the decomposition groups,

$$\mathcal{G}_{\overline{k(v)}} \longrightarrow \mathcal{G}_{\overline{K}}.$$

For (a) and (b) to define a sheaf, it is enough that the specialization maps be isomorphisms almost everywhere.

7.3. It will be convenient to modify the preceding definition by replacing Galois groups by Weil groups. Let Y be a scheme over $\mathbb{F}_p$. Let $\overline{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$, and let

$$\overline{Y} = Y \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p.$$

A Weil sheaf, respectively a constructible Weil sheaf of A -modules, $\mathcal{G}$ on Y , consists of a sheaf, respectively a constructible sheaf of A -modules, $\overline{\mathcal{G}}$ on $\overline{Y}$, together with an action of Frobenius on $\overline{\mathcal{G}}$, lying above its action on $\overline{Y}$.

Unneeded remark. Weil sheaves of sets on Y form a topos Y_{Weil} . It may be described as a 2-fiber product of topoi: identifying $\text{Spec}(\mathbb{F}_p)_{\text{ét}}$ with the classifying topos $B\mathbb{Z}$, one has

$$Y_{\text{Weil}} \simeq \overline{Y}_{\text{ét}} \times_{B\mathbb{Z}} B\mathbb{Z}.$$

We shall simply call the Weil sheaves on Y sheaves; sheaves on $Y_{\text{ét}}$ will be called étale sheaves.

7.4. A constructible A -module sheaf in the sense of 7.3 can be described in Galois terms as in 7.2, replacing everywhere

$$\text{Gal}(\overline{K}/K), \quad \text{Gal}(\overline{K}_v/K_v), \quad \text{Gal}(\overline{k}_v/k_v)$$

by

$$W(\overline{K}/K), \quad W(\overline{K}_v/K_v), \quad W(\overline{k}_v/k_v) = \mathbb{Z}.$$

7.5. Let K_v be a nonarchimedean local field. A Weil sheaf on $\text{Spec}(\mathcal{O}_v)$ is defined by paraphrasing 7.3. Let $\overline{K}_v$ be a separable closure of K_v . Giving a constructible Weil sheaf of A -modules $\mathcal{G}$ on $\text{Spec}(\mathcal{O}_v)$ is equivalent to giving:

- a) the fiber $\mathcal{G}_{\overline{K}_v}$, an A - $W(\overline{K}_v/K_v)$ -module, of finite type over A ;
- b) the fiber $\mathcal{G}_{\overline{k}_v}$, an A - $W(\overline{k}_v/k_v)$ -module, of finite type over A ;
- c) the $W(\overline{K}_v/K_v)$ -equivariant specialization map

$$\text{sp}_v : \mathcal{G}_{\overline{k}_v} \longrightarrow \mathcal{G}_{\overline{K}_v}.$$

Let v be a place of K , and put

$$X_v = \text{Spec}(\mathcal{O}_v).$$

By restriction, every constructible Weil sheaf of A -modules on X defines one on X_v .

A sheaf is flat if its fibers are flat A -modules.

7.6. Let K_v be a nonarchimedean local field. Put

$$\deg(v) = [k_v : \mathbb{F}_p],$$

and write ω_v^t for the unramified quasi-character of $W(\overline{K}_v/K_v)$, with values in $A(t)$, such that

$$\omega_v^t(F_v) = t^{\deg(v)}.$$

Let $\mathcal{G}$ be a flat constructible sheaf on $\text{Spec}(\mathcal{O}_v)$. Put

$$Z(\mathcal{G}_v, t) = \det(1 - F_v t^{\deg(v)}, \mathcal{G}_{\overline{K}_v})^{-1} = \det(1 - F_v, \mathcal{G}_{\overline{K}_v} \otimes \omega_v^t)^{-1}. \quad (7.6.1)$$

For p invertible in A , define a conductor

$$a(\mathcal{G}) = \text{sw}(\mathcal{G}_{\overline{K}_v}) + \dim(\mathcal{G}_{\overline{K}_v}) - \dim(\mathcal{G}_{\overline{k}_v}), \quad (7.6.2)$$

an integer if A is local, and in general a locally constant function on $\text{Spec}(A)$.

Assume that A is a field of characteristic $\neq p$. Let ψ be a nontrivial additive character of K_v with values in A^* , and let dx be a Haar measure with values in A , as in 6.5. Put

$$\varepsilon(\mathcal{G}_v, \psi, dx, t) = \varepsilon_0(\mathcal{G}_{\overline{K}_v} \otimes \omega_v^t, \psi, dx) \det(-F_v t^{\deg(v)}, \mathcal{G}_{\overline{k}_v})^{-1}. \quad (7.6.3)$$

This is a monomial. By (5.5.2), its degree in t is

$$a(\mathcal{G}) \deg(v).$$

Remark 7.7. Let M be a representation of $W(\overline{K}_v/K_v)$. In Galois terms, one defines sheaves $j_!M$ and j_*M on $\text{Spec}(\mathcal{O}_v)$ by the specialization maps

$$0 \longrightarrow M, \quad M^{I_v} \longrightarrow M.$$

The quantities ε and ε_0 of the preceding sections are identified with the quantity (7.6.3) relative respectively to j_*M and $j_!M$. Unlike j_* , the functor $j_!$ commutes with reduction modulo ℓ . This may explain why, in §6, we considered only ε_0 .

7.8. Let $\mathcal{G}$ be a constructible sheaf of flat A -modules on X . For every place v of K , $\mathcal{G}$ induces a sheaf $\mathcal{G}_v$ on $\text{Spec}(\mathcal{O}_v)$. Put

$$Z(\mathcal{G}, t) = \prod_v Z(\mathcal{G}_v, t) \in A[[t]]. \quad (7.8.1)$$

7.9. Assume that A is a field of characteristic $\neq p$. On the adèle ring of K there is a unique Haar measure with values in $\mathbb{Z}[1/p]$, such that

$$\int_{\mathbb{A}/K} dx = 1.$$

One can write

$$dx = \bigotimes_v dx_v$$

where dx_v is a Haar measure on K_v , with values in $\mathbb{Z}[1/p]$, and, for almost all v ,

$$\int_{\mathcal{O}_v} dx_v = 1.$$

The dx_v define Haar measures with values in A .

If there exists a nontrivial character ψ of $\mathbb{A}/K$, with values in A^* , put

$$\varepsilon(\mathcal{G}, t) = \prod_v \varepsilon(\mathcal{G}_v, \psi_v, dx_v, t). \quad (7.9.1)$$

Almost all terms in the product are equal to 1, and the product does not depend on the choice of ψ nor on the chosen decomposition of dx .

Let A' be obtained from A by adjoining a primitive p -th root of unity. There exists a ψ with values in A'^* . The corresponding global constant, being independent of ψ , is invariant under $\text{Gal}(A'/A)$, hence lies in A . This makes it possible to define ε even if no character ψ exists.

7.10. Let i denote the inclusion of $\text{Spec}(K)$ into X . If V is a representation of $W(\overline{K}/K)$, write i_*V for the sheaf on X with fiber over $\overline{K}$ equal to V , and for which the specialization maps are

$$\text{sp}_v : V^{I_v} \hookrightarrow V.$$

Theorem 7.11. Let A be a field of characteristic $\ell \neq p$. Let $\mathcal{G}$ be a constructible sheaf of A -vector spaces, and let V be the fiber of $\mathcal{G}$ over $\overline{K}$.

(i) $Z(\mathcal{G}, t)$ is a rational function of t .

(ii) As $t \rightarrow \infty$, $Z(\mathcal{G}, t)$ is asymptotic to $\varepsilon(\mathcal{G}, t)$. In other words,

$$Z(\mathcal{G}, t)/\varepsilon(\mathcal{G}, t)$$

is a formal series in t^{-1} , with constant term 1.

(iii)

$$Z(i_*V, t) = \varepsilon(i_*V, t) Z(i_*V^*, pt^{-1}).$$

Let S be a finite set of places containing all the places where the representation V is ramified. Let j be the inclusion of $X - S$ into X , and let $j_!V$ be the sheaf on X , with generic fiber V , unramified outside S , and with zero fiber at $v \in S$. One checks immediately that (i) and (ii) are equivalent to

$$Z(j_!V, t) \text{ is a rational function of } t, \quad (\text{i}')$$

$$\text{as } t \rightarrow \infty, \quad Z(j_!V, t) \sim \varepsilon(j_!V, t). \quad (\text{ii}')$$

Assume $\ell \neq 0$. For every representation W of $W(\overline{K}/K)$, unramified outside S , identified with a locally constant sheaf on $X - S$, put

$$R^0 j_* W = j_* W = i_* W,$$

and let $R^1 j_* W$ be the sheaf zero outside S , with fiber at $v \in S$ equal to $H^1(I_v, W)$. Put

$$Z(Rj_* W, t) = Z(R^0 j_* W, t) Z(R^1 j_* W, t)^{-1}. \quad (7.11.1)$$

Let P' be the kernel of

$$t_\ell : I_v \longrightarrow \mathbb{Z}_\ell(1).$$

For every representation W of I_v over A , one has canonically

$$H^i(I_v, W) = H^i(I_v/P', W^{P'}). \quad (7.11.2)$$

For every representation W of $\mathbb{Z}_\ell(1)$, one has canonically

$$H^0(\mathbb{Z}_\ell(1), W) = W^{\mathbb{Z}_\ell(1)} \quad (\text{invariants}), \quad (7.11.3)$$

$$H^1(\mathbb{Z}_\ell(1), W) = W_{\mathbb{Z}_\ell(1)}(-1) \quad (\text{twisted coinvariants}),$$

$$H^i(\mathbb{Z}_\ell(1), W) = 0 \quad (i \geq 2).$$

Since duality exchanges invariants and coinvariants, one has, cf. 2.3:

Lemma 7.11.4. For $\ell \neq 0$, $H^0(I_v, W)$ and $H^1(I_v, \omega_1 W^*)$ are canonically in duality.

We shall not use the fact that this duality can be interpreted as a cup product with values in

$$H^1(I_v, A(1)) = \text{Hom}(I_v, A(1)) \simeq A,$$

generated by t_ℓ modulo ℓ .

For $\ell = 0$, in the course of this proof, put

$$H^1(I_v, W) := H^0(I_v, \omega_1 W^*)^*,$$

and define $Z(Rj_* W, t)$ by (7.11.1). This definition is justified by 7.11.5 and 7.11.6 below. For W a representation of $W(\overline{K}_v/K_v)$, put also

$$Z_v(Rj_* W, t) = \det(1 - Ft^{\deg(v)}, H^0(I_v, W))^{-1} \det(1 - Ft^{\deg(v)}, H^1(I_v, W)).$$

Let A be a discrete valuation ring of unequal characteristic $\ell \neq p$, and let W be a free A -module on which $W(\overline{K}_v/K_v)$ acts. Let η be the fraction field of A , and $s = A/\mathfrak{m}$ its residue field.

Lemma 7.11.5.

$$Z_v(Rj_* W_s, t) = Z_v(Rj_* W_\eta, t) \pmod{\mathfrak{m}}.$$

The proof below will lose its mystery in §10. Let

$$P' = \text{Ker}(t_\ell : I \rightarrow \mathbb{Z}_\ell(1)),$$

let σ be a generator of $\mathbb{Z}_\ell(1)$, let F be a geometric Frobenius, and let $K^\bullet$ be the complex concentrated in degrees 0 and 1:

$$K^\bullet : \quad W^{P'} \xrightarrow{\sigma-1} W^{P'}.$$

The endomorphism

$$\sum_{i=0}^{q-1} \sigma^i$$

of $W^{P'}$ is invertible: it is enough to see this after reduction modulo the maximal ideal of A . After reduction, since some power σ^{ℓ^n} is the identity, σ is unipotent, and q is the unique eigenvalue of $\sum \sigma^i$.

Let $\underline{F} = (F_0, F_1)$ be the endomorphism of $K^\bullet$ with components

$$F_0 = F, \quad F_1 = F \circ \left(\sum_{i=0}^{q-1} \sigma^i \right)^{-1}.$$

The quantity

$$Z = \det(1 - F_0 t^{\deg(v)}, W^{P'})^{-1} \det(1 - F_1 t^{\deg(v)}, W^{P'})$$

is compatible with extension of scalars to η or to s . After extension of scalars to η , respectively to s , one has

$$Z = \det(1 - F t^{\deg(v)}, H^0(K^\bullet))^{-1} \det(1 - F t^{\deg(v)}, H^1(K^\bullet)).$$

One concludes by noting that

$$(H^i(K_\eta^\bullet), H^i(\underline{F})) \simeq (H^i(I_v, W_\eta), F),$$

and

$$(H^i(K_s^\bullet), H^i(\underline{F})) \simeq (H^i(I_v, W_s), F).$$

Variante 7.11.6. Suppose A is complete, and let W be an A -module on which $W(\overline{K}_v/K_v)$ acts continuously for the ℓ -adic topology, and no longer, as above, for the discrete topology. Put again

$$H^1(I_v, W_\eta) = (W_\eta^{P'})_{\mathbb{Z}_\ell(1)}(-1) \quad (\text{twisted coinvariants}),$$

and define $Z_v(Rj_*W, t)$ as before. The H^1 defined above can be interpreted as a continuous H^1 :

$$H^1(I_v, W_\eta) = \varprojlim_n H^1(I_v, W/\ell^n) \otimes_A \eta. \quad (7.11.6.1)$$

The reduction formula 7.11.5 remains valid, with the same proof.

Lemma 7.11.7. Formula (iii) is equivalent to

$$Z(j_!V, t) = \varepsilon(j_!V, t) Z(Rj_*V, pt^{-1}). \quad (7.11.8)$$

Lemma 7.11.7 follows from the local identity

$$\det(1 - F_v t^{\deg(v)}, H^0(I_v, V))^{-1} = \det(-F_v t^{\deg(v)}, H^0(I_v, V))^{-1} \det(1 - F_v (pt^{-1})^{\deg(v)}, H^1(I_v, V^*))^{-1}.$$

If V is replaced by the representation $V \otimes \omega^t$ of $W(\overline{K}_v/K_v)$ over $A((t))$, this formula simplifies to

$$\det(1 - F_v^{-1}, H^0(I_v, V)) = \det(-F_v, H^0(I_v, V)) \det(1 - F_v, H^1(I_v, \omega_1 \otimes V^*)),$$

which follows from local duality 7.11.4.

We prove 7.11. If $A = \mathbb{C}$, the infinite product $Z(j_*V, t)$ converges for $|t|$ sufficiently small, and

$$L(V, s) = Z(j_*V, q^{-s}).$$

Thus Z is a meromorphic function of t ; the functional equation shows that it has moderate growth as $t \rightarrow \infty$, hence is a rational function of t . The functional equation of the L -function gives (iii).

The statements (i) and (iii) are purely algebraic. Since they are true for $A = \mathbb{C}$, and so are (i') and (7.11.8), they remain true for A of characteristic 0. Now suppose A has characteristic $\ell \neq 0$.

Let J be an open normal subgroup of $W(\overline{K}/K)$, containing I_v for $v \in S$. If a representation V of $W(\overline{K}/K)/J$ over A lifts to characteristic 0, the assertions (i') and (7.11.8) for V are obtained by reduction modulo ℓ , taking into account, for (7.11.8), 7.11.5. The quantities

$$Z(j_! V, t), \quad Z(Rj_* V, t), \quad \varepsilon(j_! V, t)$$

depend additively on V , an extension giving a product. To obtain (i) and (7.11.8) in general, it is therefore enough to know that, after extension of scalars, every representation of $W(\overline{K}/K)/J$ virtually lifts to characteristic 0. The method of 4.10, cf. 4.10.3, reduces this question to the case of finite groups, i.e. to Brauer's theorem.

Statement (ii') follows from (7.11.8) and from the fact that, for $t = 0$, $Z = 1$.

Proposition 7.12. If A is a field of characteristic p , and $\mathcal{G}$ a constructible sheaf of A -vector spaces, then $Z(\mathcal{G}, t)$ is still a rational function of t .

In the proof of 7.11(i), we did not in fact use the hypothesis $\ell \neq p$.

8. ℓ -adic representations of nonarchimedean local Weil groups; Hodge structures and archimedean local Weil groups

8.1. Let K be a nonarchimedean local field. We use the notation of 2.1 and 2.2.

Let ℓ be a prime number $\neq p$, and let E_λ be a finite extension of $\mathbb{Q}_\ell$; more generally, one could take for E_λ a valued field extending $\mathbb{Q}_\ell$. A λ -adic representation V of $W(\overline{K}/K)$ is a representation continuous for the λ -adic topology

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

of $W(\overline{K}/K)$ on a finite-dimensional E_λ -vector space.

The following result is proved in [13].

Theorem 8.2 (Grothendieck). Let (V, ρ) be a λ -adic representation of $W(\overline{K}/K)$. There exists

$$N \in \mathrm{End}(V)(-1),$$

nilpotent, such that, for σ in a finite-index subgroup of I , one has

$$\rho(\sigma) = \exp(N t_\ell(\sigma)).$$

This $N \in \mathrm{End}(V)(-1)$ is unique, hence Galois-invariant; for $w \in W(\overline{K}/K)$, one has

$$\rho(w)N\rho(w)^{-1} = q^{v'(w)}N.$$

We shall use 8.2 to describe λ -adic representations of $W(\overline{K}/K)$ in terms independent of the topology of E_λ . This will be done twice: 8.3 is more intrinsic, and 8.4, which is enough for us, is more elementary.

8.3.1. Up to unique isomorphism, there exists a unique algebraic group G over $\mathbb{Q}_\ell$, isomorphic to $\mathbb{G}_a$, and endowed with an isomorphism

$$\mathbb{Q}_\ell(1) \xrightarrow{\sim} G(\mathbb{Q}_\ell).$$

It is denoted $\mathbb{G}_a(1)$.

8.3.2. Let $J \subset I$ be an open invariant subgroup of $W(\overline{K}/K)$, and put

$$J' = J \cap \mathrm{Ker}(t_\ell).$$

The quotient $W(\overline{K}/K)/J'$ is an extension of the discrete group $W(\overline{K}/K)/J$ by J/J' . Applying to this extension

$$t_\ell : J/J' \longrightarrow \mathbb{Z}_\ell(1) \longrightarrow \mathbb{G}_a(1),$$

one obtains an extension W_J^ℓ of $W(\overline{K}/K)/J$ by $\mathbb{G}_a(1)$, see Serre [12], II, 1.3:

$$\begin{array}{ccccccc} 0 & \longrightarrow & J/J' & \longrightarrow & W(\overline{K}/K)/J' & \longrightarrow & W(\overline{K}/K)/J \longrightarrow 0 \\ & & \downarrow t_\ell & & \downarrow \alpha & & \parallel \\ 0 & \longrightarrow & \mathbb{G}_a(1) & \longrightarrow & W_J^\ell & \longrightarrow & W(\overline{K}/K)/J \longrightarrow 0. \end{array}$$

8.3.3. The W_J^ℓ form a projective system, and 8.2 means that every λ -adic representation of $W(\overline{K}/K)$ factors through an algebraic representation of some W_J^ℓ , i.e. through a representation of the group scheme

$$W^\ell(\overline{K}/K) = \varprojlim_J W_J^\ell.$$

8.3.4. Let I_J^ℓ be the inverse image of $I/J \subset W(\overline{K}/K)/J$, and put

$$I^\ell = \varprojlim_J I_J^\ell.$$

To $\bar{\sigma} \in I/J$, the image of $\sigma \in I$, associate the element $\alpha(\sigma) t(-t_\ell(\sigma))$ of W_J^ℓ . This section defines splittings

$$I_J^\ell = \mathbb{G}_a(1) \times I/J, \quad I^\ell = \mathbb{G}_a(1) \times I.$$

There is a diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & W(\overline{K}/K) & \longrightarrow & \mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & I \times \mathbb{G}_a(1) & \longrightarrow & W^\ell(\overline{K}/K) & \longrightarrow & \mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{G}_a(1) & \longrightarrow & \mathbb{G}_a(1) & & \end{array}$$

8.3.5. Let $'W^\ell$ be the semidirect product of $W(\overline{K}/K)$ by $\mathbb{G}_a(1)$, on which $W(\overline{K}/K)$ acts by

$$wxw^{-1} = q^{v'(w)}x.$$

There exist isomorphisms β between $W^\ell(\overline{K}/K)$ and $'W^\ell(\overline{K}/K)$ making the diagram

$$\begin{array}{ccccc} I \times \mathbb{G}_a(1) & \longrightarrow & W^\ell(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) \\ \parallel & & \downarrow \beta & & \parallel \\ I \times \mathbb{G}_a(1) & \longrightarrow & 'W^\ell(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) \end{array}$$

commutative, and one checks easily that any two are conjugate by an element of $\mathbb{G}_a(1)(\mathbb{Q}_\ell)$. The indeterminacy lies in the choice of a lift of a Frobenius $F \in W(\overline{K}/K)$, and one uses $q - 1 \neq 0$; see 8.4.3 below.

8.3.6. Let $'W$ be the group scheme over $\mathbb{Q}$, the semidirect product of $W(\overline{K}/K)$ by $\mathbb{G}_a$, on which $W(\overline{K}/K)$ acts by

$$wxw^{-1} = q^{v'(w)}x.$$

Every isomorphism

$$\mathbb{G}_a \xrightarrow{\sim} \mathbb{G}_a(1), \quad \text{i.e. } \mathbb{Q}_\ell \xrightarrow{\sim} \mathbb{Q}_\ell(1),$$

defines an isomorphism

$$'W_{\mathbb{Q}_\ell} \xrightarrow{\sim} 'W^\ell,$$

hence a class of such automorphisms modulo $\mathbb{Q}_\ell^*$, acting on $'W^\ell$ through its action on $\mathbb{G}_a(1)$. One checks, see 8.4.3 below, that this automorphism group acts trivially on the set of isomorphism classes of representations of $'W^\ell$. In total one has obtained:

(8.3.7). The isomorphism classes of λ -adic representations of $W(\overline{K}/K)$ are in natural bijection with the isomorphism classes of representations over E_λ of the group scheme over $\mathbb{Q}$, $'W$.

Definition 8.4.1. Let E be a field of characteristic 0. A representation of $'W(\overline{K}/K)$ over E is a pair

$$\rho = (\rho', N')$$

consisting of:

- (a) a finite-dimensional representation

$$\rho' : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

of $W(\overline{K}/K)$ over E ;

- (b) a nilpotent endomorphism N' of V , such that

$$\rho'(w)N'\rho'(w)^{-1} = q^{v'(w)}N'. \quad (8.4.1.1)$$

8.4.2. Choose a geometric Frobenius $F \in W(\overline{K}/K)$ and an isomorphism

$$\tau : \mathbb{Q}_\ell(1) \xrightarrow{\sim} \mathbb{Q}_\ell.$$

To a λ -adic representation (V, ρ) of $W(\overline{K}/K)$ one associates a representation (ρ', N') of $'W(\overline{K}/K)$ on E_λ as follows:

- (a) For $\sigma \in I$, put, with N defined by 8.2,

$$\rho'(F^n \sigma) = \rho(F^n \sigma) \exp(-t_\ell(\sigma)N).$$

- (b) $N' \in \mathrm{End}(V)$ corresponds to N , via τ .

Lemma 8.4.3. The isomorphism class of (ρ', N') depends only on ρ , not on the choices of F and τ .

Change F to $F\varpi$, obtaining (ρ'', N') . One has

$$\rho''(F^n \sigma) = \exp(n t_\ell(\varpi)N) \rho'(F^n \sigma) = A \rho'(F^n \sigma) A^{-1},$$

where

$$A = \exp((1 - q^{-1})^{-1} t_\ell(\varpi)N).$$

Changing τ to $a\tau$ changes (ρ', N') into (ρ', aN') . We shall show that, for every representation (ρ', N') of $'W(\overline{K}/K)$ over a field E , and every $a \in E^\times$,

$$(\rho', N') \simeq (\rho', aN').$$

Let $I' = \mathrm{Ker}(\rho')$, and choose n such that F^n is central in W/I' , for example $\# \mathrm{Aut}(I/I')$. For α a conjugacy class in an algebraic closure of E , let V^α be the largest subspace of V , stable under F^n , such that the eigenvalues of $F^n|_{V^\alpha}$ lie in α .

One checks:

- (a) V^α is stable under $\rho'(W(\overline{K}/K))$, and $V = \bigoplus V^\alpha$;

- (b) $N'V^\alpha \subset V^{q^{-n}\alpha}$.

If A is the linear transformation preserving the V^α , and such that

$$A|_{V^\alpha} = \lambda(\alpha),$$

with

$$\lambda(q^{-n}\alpha) = a\lambda(\alpha),$$

then

$$A(\rho', N')A^{-1} = (\rho', aN').$$

The conclusion 8.3.7 follows at once from 8.4.3.

8.5. Let (ρ', N') be a representation of $'W(\overline{K}/K)$ over E . Let u be the unipotent component of $\rho'(F)$:

$$\rho'(F) = \rho'(F)^{ss} u.$$

Since $N' \in \text{End}(V)$ is an eigenvector for conjugation by $\rho'(F)$, u and N' commute. Since a power of F is central in $\text{im}(\rho')$, a power of u , and hence u , centralizes $\text{im}(\rho')$. If

$$w_1, w_2 \in W(\overline{K}/K) - I,$$

then $\rho'(w_1)$ and $\rho'(w_2)$ have a common power. It follows that u^n is the unipotent component of $\rho'(F^n \sigma)$, for $\sigma \in I$, and, for $n \neq 0$, also that of

$$\rho'(F^n \sigma) \exp(aN'),$$

which is conjugate to it by

$$\exp((1 - q^{-n})^{-1} aN').$$

Put

$$\rho'^{ss}(F^n \sigma) = \rho'(F^n \sigma) u^{-n}. \quad (8.5.1)$$

Definition 8.6.

- (i) (ρ'^{ss}, N') is the F -semisimplification of (ρ', N') .
- (ii) One says that (ρ', N') is F -semisimple if $\rho' = \rho'^{ss}$, i.e. if, for one, respectively for every,

$$w \in W(\overline{K}/K) - I,$$

and for one, respectively for every, $a \in E$,

$$\rho'(w) \exp(aN')$$

is semisimple.

For a λ -adic representation, its F -semisimplification is again defined by (8.5.1), and F -semisimplicity as in (ii).

8.7. The formalism of 8.3 or 8.4 makes it possible to compare λ -adic representations for different fields E_λ . To do this, introduce the following classical terminology.

Let F be an extension of E , and let ρ be a representation of $'W(\overline{K}/K)$ over F . We say that ρ is *defined over E* if, for every, or for one, algebraically closed extension $\overline{F}$ of F , $\rho_{\overline{F}}$ is isomorphic to its conjugates under $\text{Aut}(\overline{F}/E)$.

Let F_1 and F_2 be two extensions of E , and let ρ_i be a representation of $'W(\overline{K}/K)$ over F_i . We say that ρ_1 and ρ_2 are compatible if they are defined over E , and if, for a common algebraically closed extension

$$F \supset F_1, F_2 \supset E,$$

the representations $(\rho_1)_F$ and $(\rho_2)_F$ are isomorphic.

Definition 8.8. Let E be a number field, let λ_1 and λ_2 be two finite places of E not dividing p , and let ρ_1, ρ_2 be λ_1 - and λ_2 -adic representations of $W(\overline{K}/K)$. We say that ρ_1 and ρ_2 are compatible if the corresponding representations of $'W(\overline{K}/K)$ are compatible in the sense of 8.7.

We leave to the reader the verification of the following pleasant exercise.

Proposition 8.9. Suppose that (V_1, ρ_1) and (V_2, ρ_2) are F -semisimple, and write N_i for the monodromy operator attached to (V_i, ρ_i) . Let W be the unique increasing finite filtration of V_i such that

$$N_i W^k \subset W^{k-2}$$

and such that N_i^j induces an isomorphism

$$\text{Gr}_W^j(V_i) \xrightarrow{N_i^j} \text{Gr}_W^{-j}(V_i).$$

For ρ_1 and ρ_2 to be compatible, it is necessary and sufficient that the characters of the representations of $W(\overline{K}/K)$ on the $\text{Gr}_W^j(V_1)$ and $\text{Gr}_W^j(V_2)$ have values in E and be respectively equal.

Example 8.10. Take $E = \mathbb{Q}$. Let A be an abelian variety over K . The representations of

$$W(\overline{K}/K) \subset \text{Gal}(\overline{K}/K)$$

on

$$V_\ell(A) = T_\ell(A) \otimes \mathbb{Q}_\ell$$

are F -semisimple and compatible.

Variant 8.11. Let G be a linear algebraic group over E , of characteristic 0. A representation of $'W$ in G , over E , is a pair (ρ', N') with

$$\rho' : W(\overline{K}/K) \longrightarrow G(E)$$

trivial on an open subgroup, and $N' \in \text{Lie}(G)$, satisfying (8.4.1.1). For $E = E_\lambda$, the formulas (8.4.2) associate a representation of $'W$ in G to every representation, continuous for the λ -adic topology,

$$\rho : W \longrightarrow G(E_\lambda).$$

Its geometric isomorphism class, after extension of scalars to $\overline{E}_\lambda$, does not depend on the arbitrary choices made: the first part of the proof of 8.4.3 still applies; it remains to show that

$$(\rho', N') \simeq (\rho', aN').$$

For this, note that the algebraic subgroup H of G which centralizes $\text{im}(\rho')$ and sends N' to a multiple contains an element $\rho'(F^m)$, hence an element g such that

$$\text{ad}(g)(N') = q^m N'.$$

The character $\chi : H \rightarrow \mathbb{G}_m$ giving the action on N' can therefore only be surjective.

If G is defined over a number field E , one defines compatibility of λ -adic representations

$$W(\overline{K}/K) \longrightarrow G(E_\lambda)$$

as in 8.7 and 8.8. The formal discussion 8.5, 8.6 generalizes unchanged.

8.12. Local factors. Let (ρ', N') be a representation of $'W(\overline{K}/K)$ on an E -vector space V . By abuse of notation put

$$V^I = \text{Ker}(N')^{\rho'(I)};$$

this is a representation of $W(\overline{k}/k)$. Define a conductor, a local L -factor, and local constants by

$$a(V) = sw(V, \rho') + \dim V - \dim V^I, \quad (8.12.1)$$

$$Z(V, t) = \det(1 - Ft^d, V^I)^{-1}, \quad (8.12.2)$$

$$\varepsilon_0(V, t) = \varepsilon_0(\rho', t), \quad (8.12.3)$$

$$\varepsilon(V, t) = \varepsilon_0(\rho', t) \det(-Ft, V^I)^{-1}. \quad (8.12.4)$$

If (ρ', N') is defined by a λ -adic representation ρ on V , then

$$V^I = V^{\rho(I)}, \quad Z(V, t) = \det(1 - Ft, V^{\rho(I)})^{-1},$$

and

$$\varepsilon_0(V, t) = \varepsilon_0(\text{the semisimplification of } V, t).$$

The archimedean case. In the nonarchimedean case, what geometry supplies, namely ℓ -adic cohomology, are λ -adic representations, i.e. representations of $W(\overline{K}/K)$, not representations of $W(\overline{K}/K)$.

In the archimedean case similarly, geometry supplies not representations of $W(\overline{K}/K)$, but Hodge structures. Here is the rule to use in passing from Hodge structures to representations.

(A) *K complex.* Geometry supplies vector spaces V over K , endowed with a Hodge bigrading

$$V = \bigoplus V^{pq}$$

(for example, if X is an algebraic variety over K , $H^n(X, K)$ has such a decomposition). One deduces a semisimple representation ρ of

$$W(\overline{K}/K) = K^*$$

on V by putting

$$\rho(z)v^{pq} = z^{-p}\bar{z}^{-q}v^{pq} \quad (v^{pq} \in V^{pq}).$$

(B) *K real.* Geometry supplies vector spaces V over $\overline{K}$, endowed with a Hodge bigrading

$$V = \bigoplus V^{pq}$$

and an involution F_∞ such that

$$F_\infty(V^{pq}) = V^{qp}.$$

For example, if X is an algebraic variety over K , $\text{Gal}(\overline{K}/K) = \mathbb{Z}/2\mathbb{Z}$ acts on $X(\overline{K})$ and $H^n(X(\overline{K}), \overline{K})$ has such a structure. One deduces a semisimple representation ρ of $W(\overline{K}/K)$ on V by putting

$$\rho(z)v^{pq} = z^{-p}\bar{z}^{-q}v^{pq} \quad (z \in \overline{K}^* \subset W(\overline{K}/K), v^{pq} \in V^{pq}),$$

and

$$\rho(F)v^{pq} = i^{p+q}F_\infty(v^{pq}).$$

These formulas differ from those of [3], 1.12, moreover disfigured by a misprint on p. 21, 12, because at finite places in loc. cit. I used the inverse of the local class-field-theory isomorphism used here.

At finite places,

$$H^2(\mathbb{P}^1, \mathbb{Q}_\ell) \simeq \mathbb{Q}_\ell(1)$$

corresponds to the quasi-character ω_1 . Similarly,

$$H_2(\mathbb{P}^1(\overline{K}), \overline{K}),$$

a Hodge structure of type $(-1, -1)$, with $F_\infty = -1$ if $K = \mathbb{R}$, corresponds to the quasi-character ω_1 of $W(\overline{K}/K)$.

9. Compatible systems of ℓ -adic representations

To simplify the exposition, in this section we consider only representations of Galois groups, rather than representations of Weil groups. As in 7.1, let X be a smooth projective curve over $\mathbb{F}_p$, irreducible over $\mathbb{F}_p$, with function field K .

9.1. Let E_λ be a local field, a finite extension of $\mathbb{Q}_\ell$, with $\ell \neq p$. Let W be a λ -adic representation of $\text{Gal}(\overline{K}/K)$, i.e. a representation continuous for the topology of E_λ , almost everywhere unramified, of $\text{Gal}(\overline{K}/K)$ in a finite-dimensional vector space over E_λ . Since the Galois group is compact, there exists in W an invariant lattice W^0 , a lattice being a free $\mathcal{O}_\lambda$ -submodule of W such that $E_\lambda W^0 = W$. The local factors

$$Z_v(W, t) = \det(1 - F_v t^{\deg(v)}, W_v^{I_v})^{-1}$$

are therefore in $\mathcal{O}_\lambda[[t]]$, as is their product $Z(W, t)$. Let $\text{Gal}^0(\overline{K}/K)$ be the kernel of the map from $\text{Gal}(\overline{K}/K)$ to $\text{Gal}(\mathbb{F}_p/\mathbb{F}_p)$. By a theorem of Grothendieck, $Z(W, t)$ is a rational function; it is even a polynomial if

$$W^{\text{Gal}^0(\overline{K}/K)} = W_{\text{Gal}^0(\overline{K}/K)} = 0,$$

and it satisfies a functional equation

$$Z(W, t) = \varepsilon_{\text{Gr}}(W, t) Z(W^* \omega_1, t^{-1}), \quad (9.1.1)$$

where ε_{Gr} is a monomial of degree

$$\sum_v \deg(v) a(i_* W_v)$$

(8.12.1). The reader will find in §10 a summary of Grothendieck's theory.

9.2. Let E be a number field, let $\mathcal{L}$ be an infinite set of finite places of E not dividing p , and let $\mathcal{O} \subset E$ be the ring of elements of E which are λ -integral for every $\lambda \in \mathcal{L}$. We again denote by λ the prime ideal of $\mathcal{O}$ defined by λ . Suppose given, for every $\lambda \in \mathcal{L}$, a λ -adic representation V_λ of $\text{Gal}(\bar{K}/K)$. Assume that, for every place v of K , the λ -adic representations $V_{\lambda v}$ of $W(\bar{K}_v/K_v)$ are compatible in the strict sense (8.8). It would in fact suffice to know that their F -semisimplifications (8.6) are compatible; cf. also 9.8. The rational function

$$Z(V_\lambda, t) \in \mathcal{O}(t)$$

and the “constant”, i.e. the monomial,

$$\varepsilon_\lambda(V, t) = \prod_v \varepsilon(V_{\lambda v}, \psi_v, dx_v, t) \in \mathcal{O}(t)$$

are then independent of λ ; as usual, the ψ_v are the components of a nontrivial character of $\mathbb{A}/K$, and $\otimes dx_v$ is a Tamagawa measure. Denote this function and this constant by

$$Z(V, t), \quad \varepsilon(V, t).$$

Theorem 9.3. Under the preceding hypotheses,

$$Z(V, t) = \varepsilon(V, t) Z(V^*, pt^{-1}). \quad (9.3.1)$$

Since $\mathcal{L}$ is infinite, the map

$$\mathcal{O} \longrightarrow \prod_{\lambda \in \mathcal{L}} \mathcal{O}/\lambda$$

is injective: it is enough to prove 9.3.1 after reduction modulo λ , for every $\lambda \in \mathcal{L}$. For this we shall put 9.3.1 in a form compatible with reduction modulo λ , and justified modulo λ by (7.11.8).

Let S be a finite set of places of K , containing all the places where V ramifies, and let j be the inclusion of $X - S$ in X . For every λ -adic representation W , unramified outside S , put

$$\begin{aligned} Z(j, W, t) &= \prod_{v \notin S} \det(1 - F_v t^{\deg(v)}, W_v)^{-1}, \\ Z(Rj_* W, t) &= \prod_v \det(1 - F_v t^{\deg(v)}, W_v^{I_v})^{-1} / \prod_{v \in S} \det(1 - F_v t^{\deg(v)}, W_{v, I_v}(-1))^{-1}, \\ \varepsilon(j, W, t) &= \prod_{v \notin S} \varepsilon(W_v, \psi_v, dx_v, t) \prod_{v \in S} \varepsilon_0(W_v, \psi_v, dx_v, t). \end{aligned}$$

An easy calculation, already made in 7.11.7, shows that (9.3.1) is equivalent to

$$Z(j, W, t) = \varepsilon(j, W, t) Z(Rj_* W, t). \quad (9.3.2)$$

One concludes by 7.11.6 and 7.11.8.

Stability 9.4. Let F be an extension of E , and let $\mathcal{L}_F$ be the inverse image of $\mathcal{L}$. A compatible system of λ -adic representations, $\lambda \in \mathcal{L}$, defines by extension of scalars a compatible system of λ -adic representations, $\lambda \in \mathcal{L}_F$. If χ is a character of the idèle-class group with values in the roots of unity of F , the $V_\lambda \chi$ still form a compatible system, to which 9.3 applies.

Corollary 9.5. Let χ be a character of the idèle-class group with values in the roots of unity of a finite extension of E (cf. 9.4). Let $S(\chi)$, respectively $S(V)$, be the set of places of K where χ , respectively V , ramifies. Suppose

$$S(V) \cap S(\chi) = \emptyset.$$

Then

$$\varepsilon((V - \dim V[1])([\chi] - [1]), t)$$

is, by definition,

$$\varepsilon(V\chi, t)\varepsilon(V, t)^{-1}\varepsilon(\chi, t)^{-\dim V}\varepsilon([1], t)^{\dim V}$$

and therefore

$$\prod_{v \in S(V)} \chi_v(\pi_v^{a(V_v)}) \cdot \prod_{v \in S(\chi)} \det V_v(\pi_v^{a(\chi_v)}).$$

This is a corollary of 5.5.3.

Example 9.6. For $E = \mathbb{Q}$ and $\mathcal{L}$ the set of primes other than p , if F is an elliptic curve over K , the $H^1(F, \mathbb{Q}_\ell)$ form a compatible system of ℓ -adic representations. If the modular invariant j is not constant, then, for every character χ as in 9.5, the corresponding rational functions $Z(F\chi, t)$ are polynomials and satisfy 9.5.

9.7. We refer to Weil [16] for the relation which 9.5 implies, when $\dim(V) = 2$, between $Z(V, t)$ and “modular forms” on $\mathrm{GL}(2, \mathbb{A})/\mathrm{GL}(2, K)$. The most interesting case is supplied by 9.6.

More precise results are obtained by invoking Jacquet–Langlands. Let, for example, E be an elliptic curve over K , with nonconstant modular invariant. By Langlands’ dictionary, see [4], §3, for each place v of K , the representation of

$$\mathrm{Gal}(\overline{K}_v/K_v) \simeq W(\overline{K}_v/K_v)$$

on $H^1(E, \mathbb{Q}_\ell)$ defines an irreducible admissible representation π_v of $\mathrm{GL}_2(K_v)$. Since the representations $H^1(E, \mathbb{Q}_\ell)$, for varying ℓ , are compatible, π_v can be defined over $\mathbb{Q}$ and is independent of ℓ . We shall consider the corresponding complex representation. The representation π_v is infinite-dimensional, of weight 2, i.e. the center K_v^* of $\mathrm{GL}(2, K_v)$ acts by ω_2 , and almost always has a $\mathrm{GL}(2, \mathcal{O}_v)$ -invariant vector. By 9.3, 9.6, and [8], 11.5, the restricted tensor product

$$\pi(E) = \bigotimes_v \pi_v$$

appears as a direct factor in the admissible representation

$$L_{\omega_2}^0(\mathrm{GL}(2, \mathbb{A})/\mathrm{GL}(2, K))$$

of $\mathrm{GL}(2, \mathbb{A})$ on the space of locally constant functions on $\mathrm{GL}(2, \mathbb{A})/\mathrm{GL}(2, K)$ which transform under the center by the quasi-character ω_2 , have compact support modulo the center, and are cuspidal.

Theorem 9.8. Let λ, μ be places of E , and let V_λ, V_μ be λ - and μ -adic representations of $\mathrm{Gal}(\overline{K}/K)$. Let S be a finite set of places of K , and suppose that, for $v \notin S$, the semisimplifications of $V_{\lambda v}$ and $V_{\mu v}$ are compatible. Then, for every place v , the semisimplifications of $V_{\lambda v}$ and $V_{\mu v}$ are compatible.

If S is taken large enough to contain all places ramified for V_λ or V_μ , the hypothesis reduces to:

$$(*) \quad \text{for } v \notin S, \text{ the characteristic polynomials of } F_v \text{ acting on } V_{\lambda v} \text{ and } V_{\mu v}$$

belong to $E[t]$ and are equal.

Thus Theorem 9.8 is a partial answer, in the least interesting case of function fields, to question 1 of Serre [12], I, 12.

One should be careful about the following points.

- (a) The conclusion concerns only the semisimplifications of the local representations, not their F -semisimplifications. In other words, with the notation of 8.4.2, only the semisimplification of ρ' is considered, and not N .

- (b) For $\lambda = \mu$, $V_\lambda = V_\mu$, the theorem asserts that the semisimplifications of the local representations $V_{\lambda v}$ of $W(\bar{K}_v/K_v)$ are defined over E , i.e. have character with values in E .

We may, and do, suppose that S contains all places where I_v acts through an infinite group. For $v \notin S$, there is then no longer any distinction between the semisimplification of $V_{\lambda v}$ and its F -semisimplification.

Apply Grothendieck's theorem recalled in 9.1:

$$Z(V_\lambda, t) = \varepsilon_{\text{Gr}}(V_\lambda, t) Z(V_\lambda^* \omega_1, t^{-1}).$$

This identity rewrites as

$$\prod_{v \in S} Z_v(V_\lambda, t) \prod_{v \in S} Z_v(V_\lambda^*, t^{-1})^{-1} = \varepsilon_{\text{Gr}} \frac{\prod_{v \in S} Z_v(V_\lambda^* \omega_1, t^{-1})}{\prod_{v \in S} Z_v(V_\lambda \omega_1, t)}.$$

The left-hand side is in $E(T)$, and similarly for V_μ . Hence the right-hand side is in $E(T)$ and independent of λ . Decomposing the right-hand side as the product of a monomial and of a rational function equal to 1 at $t = 0$, one obtains from the local calculation of 7.11.7 that

$$\prod_{v \in S} \frac{\det(1 - F_v t^{\deg(v)}, V_{\lambda v}^{I_v})}{\det(1 - F_v t^{\deg(v)}, V_{\lambda v, I_v}(-1))}$$

is the same for V_λ and V_μ . The factors of this product are additive in $V_{\lambda v}$, an extension giving a product, by (7.11.6.1); hence they are the same for $V_{\lambda v}$ and for its semisimplification $V_{\lambda v}^{ss}$. For this latter representation, invariants and coinvariants of I_v are the same, and one obtains

$$\prod_{v \in S} \frac{\det(1 - F_v t^{\deg(v)}, V_{\lambda v}^{ss, I_v})}{\det(1 - F_v (pt)^{\deg(v)}, V_{\lambda v}^{ss, I_v})} \quad (9.8.1)$$

which is the same for V_λ and V_μ .

Let $v_0 \in S$, and twist V_λ and V_μ by a character χ of the idèle-class group which is unramified at v_0 and very ramified at the $v \in S$ other than v_0 ; χ has values in an extension of E , and for existence see 4.14. The hypotheses made on (V_λ, V_μ) are stable under twisting. For suitable χ , after twisting, the factors of 9.8.1 corresponding to $v \neq v_0$ become 1. The factor corresponding to v_0 undergoes a trivial modification, read from $\chi(F_v)$. Put

$$P_{\lambda v}(T) = \det(1 - F_v T, (V_{\lambda v}^{ss})^{I_v}).$$

Applying the invariance of (9.8.1) to $(V_\lambda \chi, V_\mu \chi)$, one finds that, for every $v \in S$, in $E(T)$,

$$\frac{P_{\lambda v}(T)}{P_{\lambda v}(qT)} = \frac{P_{\mu v}(T)}{P_{\mu v}(qT)}. \quad (9.8.2)$$

The transformation which sends a rational function R , equal to 1 at 0, to

$$R_q(T) = \frac{R(T)}{R(qT)}$$

is injective: for divisors, one has

$$\text{div}(R) = \lim_N \text{div} \left(\prod_0^N R_q(q^i T) \right)$$

(simple limit). From 9.8.2 one deduces that $P_{\lambda v}(T) \in E[T]$ and that

$$P_{\lambda v}(T) = P_{\mu v}(T). \quad (9.8.3)$$

The hypotheses imposed on V_λ and V_μ are preserved after replacing K by a finite extension K' , replacing S by its inverse image. Since every finite local extension K'_v/K_v is induced by a global extension, because $\text{Gal}(\bar{K}_v/K_v)$ is a closed subgroup of $\text{Gal}(\bar{K}/K)$, one finds that 9.8.3 also holds for the representations of $W(\bar{K}_v/K'_v)$ defined by V_λ and V_μ . Taking for K'_v the various extensions for which the semisimplifications of these representations become unramified, one obtains, for every $F \in W(\bar{K}_v/K_v)$, $F \notin I_v$,

$$\det(1 - F_v t, V_\lambda) = \det(1 - F_v t, V_\mu).$$

Thus, outside I_v , the characters of the representations V_λ and V_μ coincide.

The same arguments, retaining from 9.8.3 only equality of degrees, show that for every open subgroup J of I_v and every character χ of J ,

$$\dim((V_{\lambda v}^{ss} \otimes \chi)^J) = \dim((V_{\mu v}^{ss} \otimes \chi)^J).$$

By Brauer's theorem, this implies that

$$V_{\lambda v}^{ss}|_{I_v} \quad \text{and} \quad V_{\mu v}^{ss}|_{I_v}$$

have the same character. The characters of V_λ and V_μ therefore coincide on all of $W(\overline{K}_v/K_v)$, completing the proof.

Proposition 9.9. Let $(V_\lambda)_{\lambda \in \mathcal{L}}$ be an infinite system of λ -adic representations. Suppose that, for every, or almost every by 9.8, place v of K , the semisimplifications of the $V_{\lambda v}$ are compatible. Then, for each λ ,

$$Z_\lambda(V, t) = \varepsilon_\lambda(V, t) Z_\lambda(V^*, pt^{-1}). \quad (9.9.1)$$

Put

$$Z_\lambda^{ss}(V, t) = \prod_v \det(1 - F_v t^{\deg(v)}, (V_{\lambda v}^{ss})^{I_v})^{-1},$$

$$\varepsilon_\lambda^{ss}(V, t) = \prod_v \varepsilon(V_{\lambda v}^{ss}, \psi_v, dx_v, t).$$

A local calculation shows that (9.9.1) is equivalent to

$$Z_\lambda^{ss}(V, t) = \varepsilon_\lambda^{ss}(V, t) Z_\lambda^{ss}(V^*, pt^{-1}). \quad (9.9.2)$$

The quotient of the local factors at v of the two sides of (9.9.1), or of (9.9.2), is additive in $V_{\lambda v}$, and these quotients coincide in the semisimple case. The identity to be proved, (9.9.2), is independent of λ . One proves it modulo λ for each λ , and concludes as in 9.3.

10. Grothendieck's theory

10.1. Let X be a nonsingular projective curve over the finite field $\mathbb{F}_p$, let $\overline{\mathbb{F}}_p$ be an algebraic closure of $\mathbb{F}_p$, and let

$$\overline{X} = X \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p$$

be obtained from X by extension of scalars from $\mathbb{F}_p$ to $\overline{\mathbb{F}}_p$. Let ℓ be a prime number prime to p , and let Λ be a noetherian ring killed by a power of ℓ .

Let G be a Weil sheaf, in the sense of 7.3 and 7.4, constructible in flat Λ -modules. We again denote by G the étale sheaf on $\overline{X}$ obtained from it.

10.2. The étale cohomology groups

$$H^i(\overline{X}, G)$$

are finite-type Λ -modules. Since they need not be projective, they will be of no use to us. We shall have to use a finer object

$$R\Gamma(\overline{X}, G) \in \text{Ob } D_{\text{parf}}(\Lambda)$$

which gives rise to them. Recall that $D_{\text{parf}}(\Lambda)$ may be defined as the category of finite complexes of finite-type projective Λ -modules, the morphisms being morphisms of complexes taken up to homotopy. The object $R\Gamma(\overline{X}, G)$ is a finite complex of finite-type projective Λ -modules, concentrated in degrees 0, 1, 2 if desired, well-defined up to homotopy, and whose cohomology groups are the $H^i(\overline{X}, G)$.

By transport of structure, the Frobenius substitution

$$\varphi \in \text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$$

acts on $R\Gamma(\overline{X}, G)$: it is an endomorphism of the complex $R\Gamma(\overline{X}, G)$, well-determined up to homotopy, and inducing on cohomology the Frobenius automorphism.

10.3. For K a finite complex of projective Λ -modules and u an endomorphism of K , put

$$\det(1 - ut, K) = \prod_i \det(1 - ut, K^i)^{(-1)^i} \in \Lambda(t). \quad (10.3.1)$$

If u and v are homotopic endomorphisms, then

$$\det(1 - ut, K) = \det(1 - vt, K),$$

so that $\det(1 - ut, K)$ is well-defined for $K \in \text{Ob } D_{\text{parf}}(\Lambda)$ and $u \in \text{End}(K)$. Taking the negative of the coefficient of t in the formal power-series expansion of $\det(1 - ut, K)$, one defines the trace

$$\text{Tr}(u, K) = \sum_i (-1)^i \text{Tr}(u, K^i). \quad (10.3.2)$$

If u is a homotopy auto-equivalence, i.e. induces an automorphism on cohomology, one can find (K', u') , isomorphic in $D_{\text{parf}}(\Lambda)$ to (K, u) , with u' an automorphism of the complex K' . One then puts

$$\det(u, K) = \prod_i \det(u, K^i)^{(-1)^i}, \quad (10.3.3)$$

and this quantity is independent of the arbitrary choices made. If u^{-1} is a homotopy inverse of u , then

$$\det(1 - ut, K) = \det(-ut, K) \det(1 - u^{-1}t^{-1}, K). \quad (10.3.4)$$

If $D(K)$ is the dual of K , the complex with components

$$D(K)^i = \text{Hom}(K^{-i}, \Lambda),$$

and if $\check{u}$ is a contragredient of u , one also has

$$\det(1 - ut, K) = \det(-ut, K) \det(1 - \check{u}t^{-1}, D(K)). \quad (10.3.5)$$

A sketch of the proof of the following theorem is given in 10.8.

Theorem 10.4 (Grothendieck). One has

$$Z(G, t) = \det(1 - Ft, R\Gamma(\overline{X}, G))^{-1}. \quad (10.4.1)$$

In applications, the hypotheses imposed on G are too restrictive. One should take as coefficients an element of $D_{\text{parf}}(X, \Lambda)$, i.e. a finite complex $G^\bullet$ of Weil sheaves of Λ -modules on X , which locally for the étale topology on $\overline{X}$ is homotopic to a finite complex of constructible flat Λ -module sheaves. Using 10.3, one can define, for x a closed point of X ,

$$Z_x(G^\bullet, t) = \det(1 - F_x t^{\deg(x)}, G_x^\bullet)^{-1}.$$

Put

$$Z(G^\bullet, t) = \prod_x Z_x(G^\bullet, t).$$

One also defines $R\Gamma(\overline{X}, G^\bullet)$, whose cohomology groups are the hypercohomology groups of $\overline{X}$ with values in $G^\bullet$, and one has

$$Z(G^\bullet, t) = \det(1 - Ft, R\Gamma(\overline{X}, G^\bullet))^{-1}. \quad (10.4.2)$$

10.5. Every $G^\bullet \in D_{\text{parf}}(X, \Lambda)$ has a dual with values in the dualizing complex $\Lambda(1)$, placed in degree -2 :

$$D(G^\bullet) = R\text{Hom}_\Lambda(G^\bullet, \Lambda(1)[2]) \in D_{\text{parf}}(X, \Lambda).$$

Its i -th cohomology sheaf is the $(2 + i)$ -th local hyper-Ext of $G^\bullet$ with $\Lambda(1)$. Poincaré duality takes the form

$$R\Gamma(\overline{X}, D(G^\bullet)) = D(R\Gamma(\overline{X}, G^\bullet)). \quad (10.5.1)$$

Applying 10.3.5, one obtains the functional equation

$$Z(G, t) = \det(-Ft, R\Gamma(\overline{X}, G))^{-1} Z(D(G), t^{-1}). \quad (10.5.2)$$

10.6. Let

$$j : U \hookrightarrow X$$

be an open subset of X , the complement of a finite set S of closed points, and let $G^\bullet \in D_{\text{parf}}(U, \Lambda)$. Write

$$j_! : D_{\text{parf}}(U, \Lambda) \longrightarrow D_{\text{parf}}(X, \Lambda)$$

for the derived functor of the exact functor extension by zero, and

$$Rj_* : D_{\text{parf}}(U, \Lambda) \longrightarrow D_{\text{parf}}(X, \Lambda)$$

for the derived functor of the direct-image functor under j . The local duality theorem, proved in particular in dimension one, gives

$$D(j_! G^\bullet) = Rj_* D(G^\bullet). \quad (10.6.1)$$

For G a locally constant sheaf of projective Λ -modules on U , with dual $\check{G}$, one has

$$D(G) = \check{G}(1)[2],$$

and (10.5.2) takes the form

$$Z_x(j_! G, t) = \det(-Ft, R\Gamma(\bar{X}, j_! G))^{-1} Z(Rj_* \check{G}(1), t^{-1}). \quad (10.6.2)$$

10.7. We shall make this formula explicit. The Z -functions on both sides are products of local factors. For the left-hand side, the local factor at a closed point x of X is

$$Z_x(j_! G, t) = \begin{cases} \det(1 - F_x t^{\deg(x)}, G_x)^{-1}, & x \notin S, \\ 1, & x \in S. \end{cases}$$

For the right-hand side it is

$$Z_x(Rj_* \check{G}(1), t) = \begin{cases} \det(1 - F_x t^{\deg(x)}, \check{G}_x(1))^{-1}, & x \notin S, \\ \det(1 - F_x t^{\deg(x)}, R\Gamma(I_x, \check{G}_K(1)))^{-1}, & x \in S. \end{cases}$$

The last formula is read as follows: $\check{G}_K(1)$ defines a representation of $W(\bar{K}_x/K_x)$, to which one applies the derived functor of the functor of invariants under I . One thereby obtains a complex of projective Λ -modules on which Frobenius acts, and one performs construction 10.3. Explicitly:

(a) The functor of invariants under I is the composite of the functor of invariants under

$$P'_\ell = \text{Ker}(t_\ell) : I\text{-modules} \longrightarrow I/P'_\ell\text{-modules},$$

and the functor of invariants under $\mathbb{Z}_\ell(1)$. The first functor is exact and transforms a projective Λ -module into a projective Λ -module. For every I -module V , one has

$$R\Gamma(I, V) = R\Gamma(\mathbb{Z}_\ell(1), V^{P'_\ell}).$$

For every $\mathbb{Z}_\ell(1)$ -module W , $R\Gamma(\mathbb{Z}_\ell(1), W)$ may be computed by taking a resolution of W by coinduced modules

$$\mathfrak{F}(\mathbb{Z}_\ell(1), M),$$

the module of locally constant functions from $\mathbb{Z}_\ell(1)$ to M , on which $\mathbb{Z}_\ell(1)$ acts by $f(x) \mapsto f(\tau x)$. For σ a generator of $\mathbb{Z}_\ell(1)$, one has an exact sequence

$$0 \longrightarrow W \longrightarrow \mathfrak{F}(\mathbb{Z}_\ell(1), W) \xrightarrow{\alpha_\sigma} \mathfrak{F}(\mathbb{Z}_\ell(1), W) \longrightarrow 0,$$

with maps $m \mapsto (i \mapsto im)$ and

$$f \longmapsto (i \mapsto f(\sigma i) - \sigma(f(i))).$$

Passing to invariants, one identifies $R\Gamma(\mathbb{Z}_\ell(1), W)$ with the complex

$$W \xrightarrow{1-\sigma} W. \quad (10.7.1)$$

When σ is replaced by σ^ν , $\nu \in \mathbb{Z}_\ell^*$, the unique commutative diagram

$$\begin{array}{ccccc} \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \xrightarrow{\alpha_\sigma} & \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \longrightarrow & 0 \\ \parallel & & \downarrow \beta & & \\ \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \xrightarrow{\alpha_{\sigma^\nu}} & \mathfrak{F}(\mathbb{Z}_\ell(1), W) & \longrightarrow & 0 \end{array}$$

gives, after passing to invariants,

$$\begin{array}{ccc} W & \xrightarrow{1-\sigma} & W \\ \downarrow 1 & & \downarrow \frac{1-\sigma^\nu}{1-\sigma} \\ W & \xrightarrow{1-\sigma^\nu} & W, \end{array} \quad (10.7.2)$$

where, for n an integer converging in $\mathbb{Z}_\ell$ to ν ,

$$\frac{1-\sigma^\nu}{1-\sigma} w = \lim_n \sum_{i=0}^{n-1} \sigma^i w. \quad (10.7.3)$$

It follows that $R\Gamma(\mathbb{Z}_\ell(1), W)$ may be represented by any of the complexes (10.7.1), these complexes, as σ varies, being identified by the isomorphisms (10.7.2).

Starting from a $W(\overline{K}/K)$ -module V , Frobenius acts by transport of structure on

$$R\Gamma(I, V) = [V^{P'_\ell} \xrightarrow{1-\sigma} V^{P'_\ell}].$$

Whatever the geometric Frobenius F , and whatever the generator σ of $\mathbb{Z}_\ell(1)$, F defines

$$\begin{array}{ccc} V^{P'_\ell} & \xrightarrow{1-\sigma} & V^{P'_\ell} \\ F \downarrow & & \downarrow F \\ V^{P'_\ell} & \xrightarrow{1-\sigma^q} & V^{P'_\ell}, \end{array}$$

which, via (10.7.2), is identified with the automorphism of the complex

$$[V^{P'_\ell} \xrightarrow{1-\sigma} V^{P'_\ell}]$$

with components

$$\left(F, \frac{1-\sigma}{1-\sigma^q} \circ F \right),$$

where

$$\frac{1-\sigma}{1-\sigma^q} = \frac{1-(\sigma^q)^{1/q}}{1-\sigma^q}$$

is defined as in (10.7.3). Hence

$$\det(1 - Ft, R\Gamma(I, V)) = \det(1 - Ft, V^{P'_\ell}) \det\left(1 - \frac{1-\sigma}{1-\sigma^q} \circ Ft, V^{P'_\ell}\right)^{-1},$$

where F denotes any geometric Frobenius and σ any generator of $\mathbb{Z}_\ell(1)$.

Formula 10.6.2 is rewritten as

$$\begin{aligned} \prod_{x \notin S} \det(1 - F_x t^{\deg(x)}, G_x) &= \det(-Ft, R\Gamma(j_! G))^{-1} \\ &\cdot \prod_{x \notin S} \det(1 - F_x t^{-\deg(x)}, \check{G}_x(1))^{-1} \\ &\cdot \prod_{x \in S} \left[\det(1 - F_x t^{-\deg(x)}, \check{G}_x^{P'_\ell}(1))^{-1} \right. \\ &\quad \cdot \det\left(1 - \frac{1-\sigma}{1-\sigma^q} F_x t^{-\deg(x)}, \check{G}_x^{P'_\ell}(1)\right) \left. \right]. \end{aligned} \quad (10.7.4)$$

10.8. Passing to the projective limit, the preceding results generalize to ℓ -adic, or λ -adic, sheaves, and one obtains the statements of 9.1. The reason, cf. 10.5.2, for the simple formula (9.1.1) is the local result that, for G locally constant on

$$U \xrightarrow{j} X,$$

one has

$$j_* D(G) = D(j_! G).$$

However, the formula obtained for the constant, the determinant of $-Ft$ acting on cohomology, is not very explicit, and there is no local theory for it.

Remark 10.9. Let X be a nonsingular projective curve over an algebraically closed field k , and let V be a locally constant sheaf of free Λ -modules on $X_{\text{ét}}$. For simplicity, assume X connected of genus $g \geq 1$ and Λ local. Here are two methods for attaching to V a free Λ -module of rank 1.

(a) Write $R\Gamma(X, V)$ as a finite complex K of free Λ -modules. Then

$$\det(R\Gamma(X, V)) = \bigotimes_i \det(K^i)^{(-1)^i},$$

where $\det$ denotes the maximal exterior power, depends, up to canonical isomorphism, only on $R\Gamma(X, V)$. We write $h(V)$ for its dual.

(b) Let

$$K = \sum_i n_i P_i$$

be a canonical divisor, $\mathcal{O}(K) \simeq \Omega_X^1$, and put

$$N_K(V) = \bigotimes_i \det(V_{P_i})^{\otimes n_i}.$$

The set of positive canonical divisors forms a projective space, and the $N_K(V)$ form a necessarily trivial local system on this projective space. Thus the $N_K(V)$, as K varies, are canonically isomorphic, and one puts

$$\langle \Omega^1, V \rangle = N_K(V).$$

If (X, V) is defined over $k_0 \subset k$, i.e. is obtained by extension of scalars from (X_0, V_0) over $\text{Spec}(k_0)$, then $\text{Gal}(k/k_0)$ acts on $h(V)$ and on $\langle \Omega^1, V \rangle$ by characters $\chi_h(V)$ and $\chi_{\Omega^1}(V)$, with values in Λ^* . These constructions generalize to the λ -adic case.

Proposition 10.9.1. Assume that one of the following conditions is satisfied:

$$\Lambda = \mathbb{Q}_\lambda \quad \text{and } V \text{ becomes trivial on a finite cover of } X; \tag{a}$$

$$V \text{ has rank 1.} \tag{b}$$

Then

$$\chi_h(V) = \chi_{\Omega^1}(V) \omega_{(1-g)\dim(V)}, \tag{10.9.1}$$

where ω_n is the character giving the Galois action on $\mathbb{Z}_\ell(n)$.

A standard argument using Chebotarev reduces us to the case where k_0 is a finite field and k is its algebraic closure. One has to show that χ_h and $\chi_{\Omega^1} \omega_{(1-g)\dim V}$ take the same value on Frobenius. In case (a), the Weil group $W(\overline{K}_0/K_0)$ of X_0 acts continuously, for the discrete topology, on $V_{\overline{K}_0}$, so that 9.3 can be applied.

Compare (9.3) with (10.7.4), for $S = \emptyset$. These are functional equations relating the same Z -functions. Expressing the fact that they have the same constant gives (10.9.1). It may be more convenient to make this calculation only for virtual V of dimension 0, and to verify (10.9.1) directly for $V = \mathbb{Z}_\ell$. In case (b), one proceeds in the same way using 10.12.1 below.

It would be very interesting to make (10.9.1) more precise by defining a canonical isomorphism between

$$h(V) \quad \text{and} \quad \langle \Omega^1, V \rangle \otimes \mathbb{Z}_\ell((1-g)\dim(V)),$$

and to generalize (10.9.1) to the ramified case.

Remark 10.10. Theorem 10.4 is probably still valid for Λ a noetherian ring of characteristic p , i.e. with $p\Lambda = 0$. The required trace formula should follow from the trace formula in coherent cohomology, the Woods Hole trace formula. The proof has not, however, been written down.

10.11. Proof of 10.4. In [7], 10.4 is proved only for ℓ -adic sheaves, because the method of proof, based on the Lefschetz trace formula on $\overline{X}$ for the powers of the Frobenius endomorphism, required division by an arbitrary integer to pass from Lefschetz to (10.4.1), through the logarithmic derivative of Z .

To obtain 10.4 as stated, one must use SGA 4, XVII, 5.5, the Lefschetz trace formula on the $\mathrm{Sym}^n(X)$, and expand both sides of 10.4.1 as series, the second expanding as

$$\det(1 - Ft, R\Gamma(\overline{X}, G))^{-1} = \sum_n \mathrm{Tr}(F, \mathrm{Sym}^n(R\Gamma(\overline{X}, G))) t^n,$$

where Sym^n is the derived functor of the nonadditive functor n -th symmetric power. Moreover, the method of proof of [7], by successive fibrations by curves, reduces the trace formula to be proved to the case of curves.

Trace formula 10.11.1. Let X be a projective smooth curve over $\mathbb{F}_q$, let U be an open subset of X , and let G be a locally constant Weil sheaf of projective Λ -modules. One has

$$\sum_{x \in U(\mathbb{F}_q)} \mathrm{Tr}(F_x, G_x) = \mathrm{Tr}(F, R\Gamma(\overline{X}, j_! G)).$$

In this formula, the Frobenii are relative to $\mathbb{F}_q$, and

$$\overline{X} = X \otimes_{\mathbb{F}_q} \overline{\mathbb{F}_q}.$$

One may suppose that $\mathbb{F}_q$ is the field of constants of the function field K of X ; otherwise both sides are 0.

The method of proof presented by Grothendieck in the oral seminar SGA 5 consists in reducing to the Lefschetz formula for the number of fixed points of endomorphisms of complete curves, already proved in [17], by studying the cohomology of covers of $\overline{X}$ as representations of the group of the cover.

Let $\overline{K}$ be an algebraic closure of K , containing the function field of $\overline{X}$. The sheaf G defines a representation of $W(\overline{K}/K)$ on the projective Λ -module $G_{\overline{K}}$. The intersection J of $\mathrm{Gal}^0(\overline{K}/K)$ and the kernel of this representation defines a connected étale cover

$$f : \overline{U}' \longrightarrow \overline{U}$$

of $\overline{U}$ on which G becomes trivial. Put

$$H = W(\overline{K}/K)/J.$$

Then H is an extension of $\mathbb{Z}$ by the group H_0 of the cover $\overline{U}'/\overline{U}$, and $G_{\overline{K}}$ is a representation of H .

Let j' be the inclusion of $\overline{U}'$ in a nonsingular projective curve $\overline{X}'$. The ℓ -adic cohomology with proper support of $\overline{U}'$ is a representation of H . Standard arguments allow one to do better, and to define

$$R\Gamma_c(\overline{U}', \mathbb{Z}_\ell) = R\Gamma(\overline{X}', j'_! \mathbb{Z}_\ell) \in D_{\mathrm{parf}}(\mathbb{Z}_\ell[H_0]),$$

on which H acts semilinearly with respect to its action on $\mathbb{Z}_\ell[H_0]$, the action of $H_0 \subset H$ being the evident action. There is a canonical isomorphism

$$R\Gamma(\overline{X}, j_! G) = [R\Gamma(\overline{X}', j'_! \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} G_{\overline{K}}]^{H_0}, \quad (10.11.2)$$

compatible with the Frobenius action on both sides.

Write $R\Gamma(\overline{X}', j'_! \mathbb{Z}_\ell)$ as a finite complex of projective $\mathbb{Z}_\ell[H_0]$ -modules on which Frobenius acts. The components of this complex appear as $\mathbb{Z}_\ell[H^+]$ -modules, for H^+ the submonoid of H , extension of $\mathbb{N} \subset \mathbb{Z}$ by H_0 . We apply the following lemma to them.

Lemma 10.11.3. Let H^+ be a semidirect product of $\mathbb{N}$ by a finite group H_0 , let M be a $\mathbb{Z}_\ell[H^+]$ -module, projective of finite rank as a $\mathbb{Z}_\ell[H_0]$ -module, let Λ be a $\mathbb{Z}_\ell$ -algebra, and let V be a $\Lambda[H^+]$ -module, projective of finite rank as a Λ -module. Let χ_M and χ_V be the characters of M and V .

- (i) There exists a function χ_M^* on H , with values in $\mathbb{Z}_\ell$, such that, for $h \in H$, and $H_0(h)$ the centralizer of h in H_0 , one has

$$\chi_M(h) = \text{Tr}(h, M) = |H_0(h)| \chi_M^*(h).$$

- (ii) For $F \in H/H_0$,

$$\text{Tr}(F, (M \otimes_{\mathbb{Z}_\ell} V)^{H_0}) = \sum_{\substack{h \mapsto F \\ \text{mod } H_0\text{-conjugacy}}} \chi_M^*(h) \chi_V(h).$$

Applying Lefschetz, one computes χ_M^* for $R\Gamma(\overline{X}', j_!'\mathbb{Z}_\ell)$, virtually the difference between $R\Gamma(\overline{X}', \mathbb{Z}_\ell)$ and an elementary quantity; 10.11.2 and 10.11.3 provide the desired formula.

10.12. The abelian case, and a fanciful application. Let

$$\chi : W(\overline{K}/K) \longrightarrow \Lambda^*$$

be a quasi-character with values in Λ^* , unramified outside a finite set S of places of K . Let j be the inclusion of $U = X - S$ in X , and let $j_![\chi]$ be the rank-one sheaf on U defined by χ , extended by 0. Put

$$\varepsilon(j_![\chi], t) = \prod_{v \in S} \varepsilon_0(\chi_v \omega_v^t, \psi_v, dx_v) \prod_{v \notin S} \varepsilon(\chi_v \omega_v^t, \psi_v, dx_v).$$

Here ω^t is as in 7.6, ε_0 is defined in 6.4, and ε , for an unramified character, is defined by 3.4.3.3.

Proposition 10.12.1. One has

$$\det(-Ft, R\Gamma(j_![\chi])) = \varepsilon(j_![\chi], t).$$

This proposition makes 10.7.4 more precise.

Let H be the quotient of $W(\overline{K}/K)$ by $\text{Ker}(\chi)$. It is enough to treat the universal case where

$$\Lambda = \mathbb{Z}/\ell^n[H].$$

This case lifts to characteristic 0, and 10.12.1 is obtained by reduction.

10.13. Take for example, for Λ , the dual numbers over $\mathbb{Z}/\ell^n$:

$$\Lambda = \mathbb{Z}/\ell^n[\epsilon] \quad (\epsilon^2 = 0),$$

let α be a continuous homomorphism from the idèle-class group $\mathbb{A}^*/K^*$ into the additive group $\mathbb{Z}/\ell^n$, and put

$$\chi = 1 + \alpha\epsilon.$$

The ramification of α is automatically tame. Let S be a finite set of places outside which α is unramified. Applying 10.4 to χ and to the trivial character, and taking the ratio, one finds that

$$L_S^+(\alpha, t) = \sum_{v \in S} \alpha_v(\pi_v) \frac{t^{\deg(v)}}{1 - t^{\deg(v)}} = \sum_n t^n \sum_{\deg(v)|n} \alpha_v(\pi_v) \quad (10.13.1)$$

lies in $\mathbb{Z}/\ell^n(t)$.

For every place v of K , choose a uniformizer π_v . Adjoin to $\mathbb{Z}/\ell^n$ a primitive p -th root of unity, and let ψ be a nontrivial additive character of $\mathbb{A}/K$, with values in the p -th roots of unity. Take the quotient of (10.7.4) for χ and of (10.7.4) for the trivial character, and use (10.12.1). After simplification, one obtains the following identity. In the left-hand side, the first two infinite sums are summed by (10.13.1). All sums run over all places of K :

$$\sum_v \alpha_v(\pi_v) \frac{t^{\deg(v)}}{1 - t^{\deg(v)}} + \sum_v \alpha_v(\pi_v) \frac{q_v t^{-\deg(v)}}{1 - q_v t^{-\deg(v)}} + \sum_v \alpha_v(-1) \frac{1}{1 - t^{\deg(v)}} \quad (10.13.2)$$

$$= - \sum_v n_v \alpha_v(\pi_v) - \sum_v \sum_{x \in k_v^*} \alpha_v(x) \psi_v(\pi_v^{1+n_v} x).$$

One checks elementarily that the validity of (10.13.2) does not depend on the choice of the uniformizers π_v , recalling that, for $x \in k_v^*$, one has $(q-1)\alpha(x) = 0$. In the left-hand side, the $\alpha_v(-1)$ are of order 2, hence vanish if $\ell \neq 2$.

11. Appendix. The calculation of ε modulo roots of unity

This appendix is a variation on a theme of Lakkis and Dwork. My attention was drawn to their results by J.-P. Serre.

We shall write $a \sim b$ for the following relation between invertible complex numbers:

$$a \sim b \quad : \quad a/b \text{ is the product of a root of unity and a power of } q^{1/2}.$$

Let K be a nonarchimedean local field, and resume the notation of 2.2.1, 2.2.2, and 3.1. The Haar measure dx is assumed to satisfy

$$\int_{\mathcal{O}} dx = q^{n/2},$$

with n an integer. Let V be a complex representation, of a finite quotient, of $\text{Gal}(\overline{K}/K)$, and let V^P denote the invariants under wild inertia. This is a subrepresentation of V .

Theorem 11.1. One has

$$\varepsilon(V, \psi, dx) \sim \varepsilon(V^P, \psi, dx).$$

Lemma 11.2. Modulo the equivalence $\sim$, $\varepsilon(V, \psi, dx)$ is independent of ψ and dx . This follows immediately from 5.3 and 5.4. This lemma will allow us to abbreviate $\varepsilon(V, \psi, dx)$ to $\varepsilon(V)$.

Lemma 11.3. If the character of V is real, i.e. if V is isomorphic to V^* , then

$$\varepsilon(V) \sim 1.$$

One may suppose that $\int_{\mathcal{O}} dx = 1$ and that $n(\psi) = 0$. One then applies 5.7.1, 5.4, and 5.5.

Let L be a finite extension of K in $\overline{K}$, let W be a representation of $\text{Gal}(\overline{K}/L)$, and let $\text{Ind}(W)$ be the representation induced to $\text{Gal}(\overline{K}/K)$.

Lemma 11.4. One has

$$\varepsilon_L(W) \sim \varepsilon_K(\text{Ind}(W)).$$

Let W' be the trivial representation of the same dimension as W . For $\psi_L = \psi_K \circ \text{Tr}_{L/K}$, (5.6.1) gives

$$\varepsilon(W - W', \psi_L) = \varepsilon(\text{Ind}(W) - \text{Ind}(W'), \psi_K),$$

and one concludes by applying 11.3 to W' and to $\text{Ind}(W')$.

11.5. Let L and W be as above, let P_L be the wild inertia group of L , and regard W^{P_L} as a representation of

$$H = P \cdot \text{Gal}(\overline{K}/L),$$

trivial on P . One has

$$\text{Ind}(W)^P = \text{Ind}_H^{\text{Gal}(\overline{K}/K)}(W^{P_L}).$$

By 11.4, the theorem for W is therefore equivalent to the theorem for $\text{Ind}(W)$. By Brauer, this reduces us to the case where V has dimension one, defined by a finite-order character χ of K^* . If χ is tame, $[\chi]^P = [\chi]$, and 11.1 is trivial. If χ is wildly ramified, $[\chi]^P = 0$, and 11.1 follows from the next lemma.

Lemma 11.6. For χ wildly ramified,

$$\varepsilon(\chi, \psi, dx) \sim 1.$$

Let m be the conductor of χ , put

$$n = \left\lceil \frac{m+1}{2} \right\rceil,$$

and let $y \in K^*$ have valuation $-m - n(\psi)$, well-defined modulo $\pi^{n-m-n(\psi)}$, such that, cf. 4.16,

$$\chi(1+a) = \psi(ay) \quad (v(a) \geq n).$$

Decompose χ as $\chi = \chi_1 \chi_2$, where χ_1 has order prime to p and χ_2 has order a power of p . As in 4.16, one proves that

$$\begin{aligned} \varepsilon(\chi, \psi, dx) &= \int \chi^{-1}(x) \psi(x) dx = \int_{y(1+(\pi^{m-n}))} \chi^{-1}(x) \psi(x) dx \\ &= \chi_1(y) \int_{y(1+(\pi^{m-n}))} \chi_2^{-1}(x) \psi(x) dx. \end{aligned} \tag{11.6.1}$$

If m is even, $m = 2n$, the integrand is constant and 11.6 is clear. For m odd, $m = 2n - 1$, there are two ways to proceed.

(a) $\chi^{-1}(x)\psi(x)$ is a nondegenerate quadratic character on

$$y(1 + (\pi^{m-n}))/ (1 + (\pi^n));$$

by Weil, the sum of its values is ~ 1 .

(b) By (11.6.1), it is enough to prove that $\varepsilon(\chi_2, \psi, dx) \sim 1$. The square of this number, multiplied by a suitable power of q , belongs to a field of p^N -th roots of unity, and has valuation 1 at every place of this field, except possibly at the unique place dividing p . It is therefore a root of unity.

Corollary 11.7. Take ψ and dx such that $n(\psi) = -1$ and

$$\int_{(\pi)} dx = 1.$$

Then $\varepsilon(V, \psi, dx)$ is an algebraic integer.

Decomposing V as $V^P + W$, one sees that it is enough to treat the following two cases.

(a) V is tamely ramified. In this case, V is a sum of representations induced from one-dimensional representations of subgroups, and ε is, up to sign, a product of Gauss sums, cf. 5.10.

(b) $V^P = 0$. In this case, $\varepsilon \sim 1$, and it is enough to prove that the complex absolute value $|\varepsilon|$ is a positive power of q . This follows from 5.7.2, where $dx'/dx = q$.

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The problem mentioned in the introduction, p. 4, of proving the conclusion of 9.9 for a Λ -adic representation has been solved:

G. Laumon, *Les constantes des équations fonctionnelles des fonctions L sur un corps global de caractéristique positive*, C.R. Acad. Sci. Paris **298**, 8 (1984), 181–184.

A fine exposition of the existence theorem for local constants is given in:

J. Tate, *Local constants*, in *Algebraic Number Fields*, edited by A. Fröhlich, Academic Press, 1977, 89–137.